# Quantifying Financial Repression in Emerging Markets*

Mert Onen†

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**Abstract**

It is hard to measure financial repression intensity in emerging markets (EMs). I propose the Domestic Bond Premium (DBP), the yield spread between local currency (LC) government and AAA-rated supranational bonds in the same currency and tenor. Across 11 EMs, I show that the DBP is frequently negative and stable during financial stress, when LC government spreads over US Treasuries widen sharply. A portfolio choice model decomposes the DBP into default, liquidity, and repression components, and the empirical evidence points to repression compressing government borrowing costs relative to supranationals. The implied fiscal footprint, however, is modest. Financial repression saves governments under one percent of their overall revenue on average. Larger savings require inflation surges or severe bond market distortions, leaving repression as an unlikely remedy for high public debt.

**Keywords:** Financial Repression, Sovereign Debt, Financial Regulation, Macroprudential Policy, Fiscal Policy.
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†University College London. Email: mert.onen.24@ucl.ac.uk.

# 1 Introduction

When conventional fiscal tools prove insufficient, governments have historically turned to financial repression, a set of policies designed to generate captive demand for public debt. In practice, this demand is generated through banking and pension fund regulations, moral suasion, taxation rules, or capital controls that steer domestic investors toward sovereign bonds, allowing governments to borrow at artificially low interest rates. Reinhart and Sbrancia (2015) estimate that such policies reduced advanced economies' (AE) real debt burdens by 1 to 5% of GDP annually between 1945-1980.

The increase in global debt levels in the aftermath of the pandemic has renewed the concern that governments may once again use financial repression to contain their debt-service costs. Emerging markets (EMs), where public debt stands at its highest level in half a century (Kose et al., 2022), should be central to this debate.[1] However, there is no measure of financial repression intensity across EMs.

Conceptually, the financial repression wedge is the difference between actual government borrowing costs ($y^{\text{Gov}}$) and the borrowing costs that would prevail without captive demand ($y^{\text{Gov, no repression}}$):

$$\text{Repression wedge}_{j,t,\tau} \equiv y^{\text{Gov}}_{j,t,\tau} - y^{\text{Gov, no repression}}_{j,t,\tau},$$

where $j$ indexes the currency, $t$ the date, and $\tau$ the tenor. As such, a negative wedge indicates that actual government borrowing costs are compressed relative to the unrepressed benchmark. However, the latter is unobserved, therefore measuring this wedge requires a proxy for the counterfactual.

This paper uses AAA-rated supranational bonds issued in EM local currencies as that proxy.[2] These bonds provide a reasonable benchmark because they can be matched to government bonds in currency, tenor, and coupon structure. At the same time, they are issued by highly creditworthy international institutions that lie outside the purview of domestic financial regulation. Given this, I define the Domestic Bond Premium (DBP) as

$$DBP_{j,t,\tau} \equiv y^{\text{Gov}}_{j,t,\tau} - y^{\text{Sup}}_{j,t,\tau},$$

where $y^{\text{Sup}}_{j,t,\tau}$ represents the yield on a supranational bond in the same currency and tenor as the government bond. Because the two yields have identical currency exposure, the DBP nets out foreign exchange risk by construction. After accounting for sovereign default risk and differential liquidity, a lower DBP indicates that government borrowing costs are compressed relative to comparable default-free benchmarks.[3]

Turkey, which I examine in detail in Section 5.5, provides a striking illustration of how negative the DBP can become. During the episode shaded in Figure 1, the authorities introduced

[1] In 2025 alone, emerging economies faced an $8.2 trillion refinancing need (Institute of International Finance, 2025). Domestic banks are already heavily exposed to sovereign debt, with government securities accounting for 17% of their total assets in the aftermath of the pandemic (International Monetary Fund, 2022).

[2] Supranational financial institutions are multilateral development banks owned and funded by several governments to finance development projects globally. For details on supranational issuance in EM currencies, see Dao and Gourinchas (2026).

[3] Throughout this paper, the term 'default-free' refers to the absence of nominal repayment, or outright default, risk. This is due to the perfect credit rating of supranational issuers in my sample.

Figure 1: Five-year Turkish lira government and supranational zero-coupon yields




*Notes:* This figure plots five-year Turkish lira denominated zero-coupon rates for government bonds (blue) and supranational bonds (orange) in percent. The shaded area marks Turkey's most acute financially repressive episode, from the December 2021 introduction of FX-protected lira deposits (KKM) to the May 2024 removal of the Securities Maintenance Practice (SMP); the green dashed lines bracket the subperiod when SMP requirements were most stringent (May 2022–June 2023). Sources: Bloomberg, Refinitiv, and author's calculations.

a sequence of financial repression policies, including restrictions on banks' foreign currency operations and mandates to hold government securities. Over the same period, annual inflation exceeded 80% and sovereign CDS spreads rose above 800 basis points. Both developments should have pushed government yields up. Instead, five-year government yields fell while lira supranational yields remained high, pushing the DBP to around $-20$ percentage points. Financial repression is the most plausible explanation for this divergence. Although Turkey is the most extreme case in the sample, it is not unique. Negative or declining DBPs amid macroeconomic stress are common across EMs.

To document this more systematically, I construct a novel dataset of supranational zero-coupon yield curves for 11 EM currencies using data from 4,017 supranational bonds issued between 1997 and 2025 by eight supranational financial institutions. Across this panel, the DBP is persistently low, frequently negative, and tends to decline during times of financial distress, meaning EM governments in my sample are often able to borrow at or even below default-free benchmarks in their own currencies despite adverse economic conditions.

To formalise the financial repression interpretation, Section 2 presents a portfolio choice model in which domestic investors are required to hold a minimum fraction of their wealth in government bonds. When the constraint binds, forced demand props up government bond prices and opens a financial repression wedge in yields. The model decomposes the DBP into default risk, liquidity, and financial repression components. Moreover, when foreign investors also hold government bonds, I show that tightening financial repression can raise domestic ownership and compress the DBP at the same time. Conditional on default and relative liquidity compensation,

this then corresponds to a larger repression wedge.

Importantly, repression has two effects that pull in opposite directions. Forced demand raises government bond prices, but when the policy is expected to persist, it may also distort the marginal value of investors' wealth and with it the shadow risk-free rate in the economy. The net effect of repression on the level of government yields is therefore theoretically ambiguous. The DBP resolves this identification problem. Because both the government and supranational bonds are denominated in the same currency, the risk-free rate component cancels in the spread; conditional on observable proxies for default risk and liquidity, movements in the DBP within a currency and tenor hence trace movements in repression intensity. The framework also disciplines the empirics with sign predictions. Rising default risk should widen the DBP, deteriorating supranational liquidity should compress it, and intensifying financial repression should compress it further, potentially below zero.

The aggregate patterns in DBP are consistent with these theoretical predictions. In addition to being persistently low and frequently negative, the DBP exhibits stability during stress episodes such as the GFC or the Covid-19 recession, when EM LC government spreads over US Treasuries surge. This insensitivity to deteriorating financial conditions is particularly pronounced at shorter maturities, where the DBP declines or turns negative during crises.

Beyond these aggregate patterns, I provide four pieces of evidence tying the DBP to financial repression. First, the variation in the DBP is predominantly country-specific. A variance decomposition of the panel dataset shows that between 80-90% of the DBP variation stems from persistent country characteristics and time-varying local factors, contrasting with the strong commonality found in foreign-currency yields (González-Rozada and Yeyati, 2008; Du and Schreger, 2016) or credit default swap (CDS) spreads (Longstaff et al., 2011). The idiosyncratic nature of the variation in DBP points to local financial conditions and domestic policies rather than global risk sentiment as the primary drivers.

Second, in panel regressions, proxies for financial repression are statistically and economically significant in explaining DBP movements across all specifications. Sovereign default risk, by contrast, is significant only at longer maturities, consistent with the low probability of outright nominal default over short horizons. Interacting default risk with domestic ownership reveals that the positive association between CDS spreads and the DBP is substantially attenuated when the domestic investor base is large, consistent with captive demand insulating government borrowing costs from deteriorating creditworthiness. These results are robust to currency–tenor–year fixed effects that absorb slow-moving unobservables such as market segmentation and capital account openness, and to replacing the domestic holdings share with reserve requirement ratios, a direct policy lever that can tilt bank portfolios toward domestic government debt.

Third, the DBP responds at high frequency to discrete policy actions. Using daily local projections around macroprudential policy events extracted from the IMF's Integrated Macroprudential Policy database (iMaPP), I find that the DBP compresses following policy events predicted to expand captive demand for LC government bonds and widens following those predicted to reduce it. Meanwhile, it is largely unresponsive to events that are not expected to affect domestic demand for government debt.

Fourth, the two narrative case studies in Section 5.5 illustrate how financial repression can

work through different mechanisms. In Turkey, shown in Figure 1, repression was deliberately imposed to control borrowing costs through banking regulation. In Russia, an exogenous geopolitical shock made financial repression exceptionally effective. Following the war in 2022, international sanctions drove out foreign investors while capital controls trapped domestic institutions in LC government bonds, holding government yields well below supranational rates despite a substantial rise in Russian sovereign default risk.

Together, these results support the use of DBP as a measure for tracking financial repression intensity across countries. Furthermore, because the DBP is built from bond prices, it is observed daily and involves no subjective classification of policies. This makes it usable in a wide range of applications.

As such, the last part of the paper asks how much governments actually gain from financial repression. Multiplying the repression component of the DBP by the outstanding stock of nominal LC government bonds converts the price distortion into an annualised fiscal saving. I then compare these savings with those from negative real interest rates, which arise when inflation erodes the real value of the same debt stock. In contrast to the earlier literature (e.g., Giovannini and de Melo, 1993; Reinhart and Sbrancia, 2015), I find that both channels have a modest fiscal footprint. Savings from captive demand stay below one percent of government revenue even at their peak. Negative real rates save more, reaching 2.4 percent of revenue for the average country at the peak of post-pandemic inflation, but they generate no revenue when inflation is stable. Repression is thus clearly visible in relative bond prices but second-order as a source of fiscal relief. This makes it an unlikely long-term remedy for high public debt, since sizable repression revenues require large bond market distortions that may lead to adverse macroeconomic consequences.

**Relation to literature.** This paper relates to several strands of literature. First, I build on the work of Du and Schreger (2016) who quantify the non-currency risk premium in EM LC sovereign yield spreads; these authors examine the yield spread between EM LC sovereign bonds and synthetic risk-free rates, defined as the US Treasury yield plus the relevant cross-currency swap rate. However, their approach also incorporates the US Treasury premium and CIP deviations.[4] Importantly, while Du and Schreger (2016) were the first to measure the government-supranational spread for two countries using limited supranational bond data, a systematic EM-wide analysis of this spread and its relation to financial repression has been absent from the literature.

Second, the finding that DBPs are frequently negative connects the paper to the safe-asset demand and convenience yield literature. Convenience yields arise when investors accept lower returns on certain assets in exchange for non-pecuniary benefits such as exceptional liquidity, universal acceptance as collateral, and safe-haven properties. Krishnamurthy and Vissing-Jorgensen (2012), Greenwood et al. (2015), Nagel (2016), Du et al. (2018), and Jiang et al. (2021) document how US Treasury securities command such premiums over comparable assets

[4]Covered Interest Parity (CIP) is a no-arbitrage condition requiring that interest rate differentials between two currencies equal the forward premium. In other words, borrowing in one currency, converting to another, and hedging exchange rate risk via forward contracts should yield the same return as borrowing directly in the second currency. Deviations from CIP may reflect excess demand for dollar hedging or limited supply of dollar forwards (Du and Schreger, 2022).

or over what standard asset-pricing models would predict. A negative DBP shows that EM LC government bonds can also trade at a premium to default-free assets. While these negative spreads superficially resemble the convenience yields documented for US Treasuries, they are more likely to arise from EM governments' ability to generate inelastic domestic demand for their liabilities. This has implications for financial stability. Unlike safe-haven assets, demand for EM government debt can drop rapidly if domestic regulations loosen or the banking sector comes under stress.[5]

In this respect, this paper contributes to the growing literature on financial repression. The term originates with McKinnon (1973) and Shaw (1973), who characterised how policies such as interest rate ceilings and reserve requirements channel domestic savings toward the government at below-market rates, with adverse consequences for capital accumulation. Giovannini and de Melo (1993) provided static estimates of the implicit tax rate that governments impose on their debtholders by suppressing local interest rates below foreign rates across 24 developing economies. Reinhart and Sbrancia (2015) document how AE sovereigns extracted resources from domestic savers through negative real interest rates alongside financial repression, showing that after World War II such policies reduced interest expense burdens significantly.

In the modern era, Becker and Ivashina (2018) and Ongena et al. (2019) provide evidence of financial repression during the European debt crisis, where moral suasion led banks to increase government bond holdings despite deteriorating sovereign creditworthiness. Broner et al. (2014) document that sovereign bond markets in turbulent times are characterised by creditor discrimination; domestic investors that are subject to local regulations absorb an increasing share of government debt as foreign investors withdraw amid rising credit risk. This mechanism is central to the theoretical framework developed in Section 2.6, where an increase in the financial repression premium on government debt can go hand-in-hand with a rise in domestic ownership share.

On the theory side, Reis (2021a) shows that financial repression can expand fiscal capacity through a wedge between the return on private capital and the interest rate on government debt, but that this comes at the cost of misallocating resources away from productive investment. In related work, Reis (2021b) models macroprudential policy as the share of government bonds that banks are required to hold and shows that tighter regulation raises bond prices along a downward-sloping demand curve, leaving a 'fiscal footprint' that lowers government borrowing costs; he stresses that once a fiscally constrained authority exploits this footprint, the line separating macroprudential policy from financial repression becomes a thin one. Chari et al. (2020) show that financial repression, though not optimal under commitment, can serve as a costly credibility mechanism. When governments force banks to hold public debt, they make a sovereign default more damaging to the domestic financial system and can thereby sustain higher debt levels. More recently, Itskhoki and Mukhin (2025) study financial repression in the currency market, comparing it with conventional policies such as FX interventions. They show that although repression can be used to reallocate resources and extract fiscal surplus under international isolation, it is generally welfare-reducing as a stabilisation tool.

[5]For a discussion of how rapidly unwinding financial repression can trigger crises, see Diaz-Alejandro (1985) on Latin America. For the sovereign-bank nexus, see Acharya et al. (2014), Gennaioli et al. (2014), and Farhi and Tirole (2018).

As Kose et al. (2022) note, EMs face increasingly limited options to manage their rapidly rising public debt levels and interest burdens since the pandemic. This constrained fiscal environment makes it increasingly relevant to study financial repression. A natural starting point is to test whether repression is systematically reflected in the pricing of EM local-currency sovereign bonds across countries and over time. The DBP provides a high-frequency, market-based measure for that purpose.

The rest of the paper proceeds as follows. Section 2 develops the theoretical framework and decomposes the DBP into default risk, relative liquidity, and financial repression components. Section 3 describes the data and construction of supranational yield curves. Section 4 presents stylised facts. Section 5 presents the empirical evidence tying repression to the DBP. Section 6 quantifies the fiscal savings that repression generates. Section 7 concludes.

# 2 Framework

This section introduces a consumption-based asset pricing model that decomposes bond yields and the DBP into interpretable components. These decompositions provide a theoretical basis for the empirical analysis in Section 5.

## 2.1 Environment and timing

Time is discrete and denoted by $t = 0, 1, 2, \ldots$ There are two types of representative investors: domestic $D$, and foreign, $F$. Both types can buy one-period LC-denominated government and supranational bonds. Let $B_{t+1}^{D,b}$ and $B_{t+1}^{F,b}$ denote purchases at date $t$, where $b \in \{Gov, Sup\}$.

A government bond with face value one trades at $P_t^{Gov}$ and repays $1 - \delta_{t+1}$ units of LC at $t+1$. The haircut $\delta_{t+1} \in [0, 1]$ is the fraction of face value lost through sovereign default between periods $t$ and $t+1$. A supranational bond trades at $P_t^{Sup}$ and repays one unit of LC with certainty. Agents incur a per-unit transaction cost $\varphi_t^b \geq 0$ when they purchase a bond. Bonds are held to maturity. There is neither trading before maturity nor a redemption fee.

Domestic agents consume $c_t$ units of a good whose LC-denominated price is $p_t > 0$ and receive a constant real endowment $\omega > 0$. Consumption and the endowment are measured in terms of goods. Bond prices and transaction costs are measured in nominal LC terms.

For domestic investors, holding either type of bond provides private convenience services (e.g., use as collateral). These services are a function of total bond holdings in a given currency. In addition, domestic investors face a regulatory constraint that compels them to hold a minimum share of their wealth in government bonds each period. This constraint reflects policies that generate inelastic domestic demand for sovereign debt (e.g., eligibility criteria for central bank liquidity facilities, treatment of government bonds in capital requirements, tax advantages, and other regulations that favour sovereign debt), and generates a financial repression wedge in government bond yields by generating captive demand.

Foreign investors can hold both bonds but receive no private convenience services and are not subject to financial repression. They value future repayments using an externally given discount factor and face portfolio management costs that increase with their holdings of each bond. These costs limit their willingness to maintain larger positions.

At the beginning of each period, both groups observe current conditions and redeem payments on their maturing bonds. Domestic investors choose consumption and new holdings of government and supranational bonds, subject to their budget and regulatory constraints. Foreign investors choose their holdings of the same two assets to maximise their next-period payoffs net of transaction and portfolio costs. Both investors take prices as given, and prices adjust to clear the bond markets.

## 2.2 Investors' portfolio choice

### 2.2.1 Domestic investors

Domestic investors choose consumption and nonnegative bond holdings to maximise

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[ u(c_t) + v(T_{t+1}) \right], \qquad T_{t+1} = B_{t+1}^{D,Gov} + B_{t+1}^{D,Sup}, \tag{1}$$

where $0 < \beta < 1$, $u' > 0$, $u'' < 0$, $v' > 0$ and $v'' < 0$. The function $v$ captures private convenience services, such as the use of bonds as collateral in interbank markets. These services depend on total face value of holdings in a given currency, with a unit of either bond contributing equally. Government bonds have an additional regulatory use, giving them a higher overall non-pecuniary service flow than supranational bonds. Per-period nominal wealth consists of the endowment and proceeds from maturing bonds:

$$W_t = p_t \omega + (1 - \delta_t) B_t^{D,Gov} + B_t^{D,Sup}. \tag{2}$$

The investor faces the budget constraint

$$p_t c_t + (P_t^{Gov} + \varphi_t^{Gov}) B_{t+1}^{D,Gov} + (P_t^{Sup} + \varphi_t^{Sup}) B_{t+1}^{D,Sup} = W_t \tag{3}$$

and the financial repression constraint

$$P_t^{Gov} B_{t+1}^{D,Gov} \geq \kappa_t W_t, \qquad \kappa_t \in [0, 1]. \tag{4}$$

The financial repression constraint applies to wealth before current consumption and saving decisions. A higher $\kappa_t$ requires the investor to allocate more resources to government bonds.

The consumption price $p_t$, sovereign haircut $\delta_t$, transaction costs $\varphi_t^{Gov}, \varphi_t^{Sup}$, and repression intensity $\kappa_t$ follow an arbitrary joint stochastic process that is unaffected by investors' choices. At each date $t$, current values are realised and observed before investors make their decisions. Future values have not yet been realised, and agents form rational expectations about them. Let $\lambda_t \geq 0$ be the multiplier on the financial repression constraint. The marginal utility of one unit of LC is $u'(c_t)/p_t$. The constrained pricing kernel and financial repression wedge are

$$\tilde{M}_{t+1} \equiv \beta \, \frac{u'(c_{t+1})/p_{t+1} - \lambda_{t+1}\kappa_{t+1}}{u'(c_t)/p_t}, \qquad \rho_t \equiv \frac{p_t \lambda_t}{u'(c_t)} \geq 0. \tag{5}$$

The marginal private convenience value, expressed in nominal LC units, is

$$\chi_t \equiv \frac{p_t v'(T_{t+1})}{u'(c_t)}. \tag{6}$$

For given levels of consumption and consumption prices, $\chi_t$ decreases with total domestic LC holdings $T_{t+1}$. Then, the domestic investor's first-order conditions imply the following Euler equations (see Appendix A for derivations):

$$P_t^{Sup} + \varphi_t^{Sup} = \mathbb{E}_t\big[\tilde{M}_{t+1}\big] + \chi_t, \tag{7}$$

$$P_t^{Gov} + \varphi_t^{Gov} = \mathbb{E}_t\big[\tilde{M}_{t+1}(1-\delta_{t+1})\big] + \chi_t + \rho_t P_t^{Gov}. \tag{8}$$

Equation (8) highlights the role of captive demand. Keeping other components fixed, when the minimum-holdings constraint binds ($\lambda_t, \rho_t > 0$), financial repression raises the price of government bonds.

### 2.2.2 Foreign investors

Foreign investors do not receive private convenience services from bond holdings and they are not subject to financial repression. In addition to the transaction costs, they incur a quadratic portfolio management cost for each bond type

$$\frac{\psi_t^{Gov}(B_{t+1}^{F,Gov})^2}{2} \text{ and } \frac{\psi_t^{Sup}(B_{t+1}^{F,Sup})^2}{2}, \text{ where } \psi_t^{Gov}, \psi_t^{Sup} > 0. \tag{9}$$

These costs represent restrictive investment mandates or funding frictions that make it less attractive to take large positions in EM domestic bond markets. Let $M_{t+1}^F$ be the foreign investors' pricing kernel, which discounts future LC repayments into current LC value. I take $M^F$ and $\psi$ as given and also do not model how gains and losses on domestic bonds affect foreigners' wealth. Then, because bonds mature after one period and portfolio costs depend only on current holdings, foreign investors choose their positions separately at each date to maximise the net payoff function

$$\begin{aligned}\max_{B_{t+1}^{F,Gov}, B_{t+1}^{F,Sup} \geq 0} \; & \mathbb{E}_t\big[M_{t+1}^F(1-\delta_{t+1})\big]B_{t+1}^{F,Gov} + \mathbb{E}_t\big[M_{t+1}^F\big]\, B_{t+1}^{F,Sup} - \big(P_t^{Gov} + \varphi_t^{Gov}\big)B_{t+1}^{F,Gov} \\ & - \big(P_t^{Sup} + \varphi_t^{Sup}\big)B_{t+1}^{F,Sup} - \frac{\psi_t^{Gov}}{2}\big(B_{t+1}^{F,Gov}\big)^2 - \frac{\psi_t^{Sup}}{2}\big(B_{t+1}^{F,Sup}\big)^2.\end{aligned} \tag{10}$$

The two positions enter the problem separately, so the first-order conditions are

$$P_t^{Gov} + \varphi_t^{Gov} + \psi_t^{Gov} B_{t+1}^{F,Gov} = \mathbb{E}_t\big[M_{t+1}^F(1-\delta_{t+1})\big], \tag{11}$$

$$P_t^{Sup} + \varphi_t^{Sup} + \psi_t^{Sup} B_{t+1}^{F,Sup} = \mathbb{E}_t\big[M_{t+1}^F\big]. \tag{12}$$

Define foreign investors' fundamental valuations net of transaction costs as

$$V_t^{F,Gov} \equiv \mathbb{E}_t\big[M_{t+1}^F(1-\delta_{t+1})\big] - \varphi_t^{Gov}, \qquad V_t^{F,Sup} \equiv \mathbb{E}_t\big[M_{t+1}^F\big] - \varphi_t^{Sup}. \tag{13}$$

Then foreign demand for domestic bonds is linear in market prices,

$$B_{t+1}^{F,Gov} = \frac{V_t^{F,Gov} - P_t^{Gov}}{\psi_t^{Gov}}, \qquad B_{t+1}^{F,Sup} = \frac{V_t^{F,Sup} - P_t^{Sup}}{\psi_t^{Sup}}. \tag{14}$$

Hence, positive $\psi_t^{Gov}$ and $\psi_t^{Sup}$ make foreign demand in each market finite and downward sloping.

### 2.3 Bond market clearing and equilibrium

Let $\bar{B}_t^{Gov}$ and $\bar{B}_t^{Sup}$ denote the exogenous quantities of bonds issued at $t$ and maturing at $t+1$. Both markets must clear each period:

$$B_{t+1}^{D,Gov} + B_{t+1}^{F,Gov} = \bar{B}_t^{Gov} \qquad \text{and} \qquad B_{t+1}^{D,Sup} + B_{t+1}^{F,Sup} = \bar{B}_t^{Sup}. \tag{15}$$

Given initial holdings and the processes $\{\delta, \varphi^{\text{Gov}}, \varphi^{\text{Sup}}, \kappa, p, \psi^{\text{Gov}}, \psi^{\text{Sup}}, M^F, \bar{B}^{Gov}, \bar{B}^{Sup}\}$, a rational-expectations equilibrium consists of bond prices, domestic consumption, and domestic and foreign bond holdings such that each investor solves its optimisation problem taking prices as given, and bond markets clear as in (15).

### 2.4 Bond yields

I focus on the case when both investor types hold both bonds. In this case, equilibrium prices satisfy the domestic and foreign optimality conditions simultaneously. I use the domestic conditions to express bond prices in terms of default compensation, transaction costs, private convenience, and a repression premium.

Although I use the domestic Euler equations, note that the terms entering these decompositions are determined jointly with foreign investment decisions through the market clearing condition (15). Define the shadow risk-free rate as $\tilde{r}_t \equiv -\log \tilde{q}_t$, where $\tilde{q}_t \equiv \mathbb{E}_t\big[\tilde{M}_{t+1}\big]$. Then, using (7)–(8) and a log-linear approximation around a frictionless, default-free benchmark, the continuously compounded bond yields are given by

$$y_t^{Gov} \approx \tilde{r}_t + \underbrace{\frac{\mathbb{E}_t[\tilde{M}_{t+1}\delta_{t+1}]}{\tilde{q}_t}}_{I_t^{Gov}} + \underbrace{\frac{\varphi_t^{Gov}}{\tilde{q}_t}}_{L_t^{Gov}} + \underbrace{\Big(-\frac{\chi_t}{\tilde{q}_t}\Big)}_{C_t^{Gov}} - \rho_t, \tag{16}$$

$$y_t^{Sup} \approx \tilde{r}_t + \underbrace{\frac{\varphi_t^{Sup}}{\tilde{q}_t}}_{L_t^{Sup}} + \underbrace{\Big(-\frac{\chi_t}{\tilde{q}_t}\Big)}_{C_t^{Sup}}. \tag{17}$$

The default risk term $I$ compensates investors for expected losses from sovereign haircuts. The liquidity term $L$ reflects compensation for trading costs. $C$ reflects the private convenience services on LC bonds. Finally, $\rho_t$ is the financial repression premium. Therefore, when the repression constraint binds ($\lambda, \rho > 0$), government bonds can trade at a premium relative to default-free but otherwise similar assets denominated in the same currency.

Given these decompositions, a natural question is whether the government yield alone is sufficient to empirically test the existence of a financial repression premium in EM LC government

bonds. Based on the model, one cannot separately identify the financial repression premium from government yields alone, because repression intensity $\kappa$ does not only impact $\rho$, but also the shadow risk-free rate $\tilde{r}$ of the domestic agents, which is unobserved. Note that

$$\tilde{r}_t = -\log\left(\mathbb{E}_t[\tilde{M}_{t+1}]\right) = -\log\left(\beta\,\mathbb{E}_t\left[\frac{u'(c_{t+1})/p_{t+1} - \lambda_{t+1}\kappa_{t+1}}{u'(c_t)/p_t}\right]\right). \tag{18}$$

Hence, if the government increases the minimum-holdings requirement at time $t$, and the policy is expected to persist, the shadow risk-free rate at time $t$ can increase. In that case, the overall direction of government bond prices will be ambiguous. Similarly, the $\chi_t$ term is unobserved and it moves with the equilibrium asset allocation. Therefore, to test the sensitivity of government borrowing costs to financial repression, $\tilde{r}_t$ and $\chi_t$ in (16) must be accounted for. The next section shows that the DBP is a useful device to account for these unobserved components.[6]

### 2.5 Domestic Bond Premium

Let $y_{j,t}^{Gov}$ and $y_{j,t}^{Sup}$ denote the zero-coupon yields at time $t$ on a government bond and a AAA-rated supranational bond in currency $j$. The DBP is defined as

$$DBP_{j,t} \equiv y_{j,t}^{Gov} - y_{j,t}^{Sup}. \tag{19}$$

Using the yield decompositions (16)–(17), and noting that the shadow risk-free rate and private convenience terms cancel out,

$$DBP_{j,t} \approx \underbrace{\frac{\mathbb{E}_t[\tilde{M}_{t+1}\delta_{j,t+1}]}{\tilde{q}_{j,t}}}_{I^{Gov}} + \underbrace{\frac{\varphi_{j,t}^{Gov} - \varphi_{j,t}^{Sup}}{\tilde{q}_{j,t}}}_{\equiv \widehat{L}^{GS}} - \rho_{j,t}. \tag{20}$$

To be able to test the predictions of the model, I need empirical proxies for the terms in (20). Sovereign default risk can be proxied by CDS premia and the relative illiquidity term can be proxied by bid–ask spreads. The financial repression wedge $\rho_t$, however, is still unobserved. To connect this component to the empirical analysis, I next examine how stronger regulatory demand can produce both a larger domestic ownership share and a larger repression wedge.

### 2.6 Domestic ownership share and the repression wedge

Let $s_t^D \equiv B_{t+1}^{D,Gov}/\bar{B}_t^{Gov}$ and $s_t^F \equiv B_{t+1}^{F,Gov}/\bar{B}_t^{Gov}$ denote government bond ownership shares of domestic and foreign investors, respectively, with $s_t^D + s_t^F = 1$. Using (14) and (15), the equilibrium government bond price satisfies

$$P_t^{Gov} = V_t^{F,Gov} - \psi_t^{Gov} s_t^F \bar{B}_t^{Gov}. \tag{21}$$

[6] An alternative measure is the local currency credit spread (LCCS) of Du and Schreger (2016), which compares LC government yields with currency-hedged US Treasuries. However, the two legs of this spread are in different currencies. Therefore, LCCS retains the deviations from covered interest parity and Treasury convenience yields, both of which reflect global conditions rather than domestic policy. Appendix B provides a conceptual and empirical comparison of the two measures.

Define the domestic valuation of government bond repayments and private convenience, net of transaction costs and excluding the current regulatory benefit, as

$$V_t^{D,Gov} \equiv \mathbb{E}_t\big[\tilde{M}_{t+1}(1-\delta_{t+1})\big] + \chi_t - \varphi_t^{Gov}. \tag{22}$$

Using the domestic Euler equation (8) and combining it with (21) then gives the equilibrium relationship:

$$\rho_t = 1 - \frac{V_t^{D,Gov}}{P_t^{Gov}} = 1 - \frac{V_t^{D,Gov}}{V_t^{F,Gov} - \psi_t^{Gov} s_t^F \bar{B}_t^{Gov}} \geq 0, \tag{23}$$

which connects the domestic ownership share to the financial repression premium.

Note that when the repression constraint binds, the market value of domestic government holdings equals $\kappa W$. Combining this condition with (21) gives

$$s_t^D \bar{B}_t^{Gov}\left[V_t^{F,Gov} - \psi_t^{Gov}(1-s_t^D)\bar{B}_t^{Gov}\right] = \kappa_t W_t. \tag{24}$$

Equation (24) determines the domestic ownership share for given regulation, wealth, government bond supply and foreign demand parameters. Now, consider a small increase in $\kappa_t$, holding all else other than current prices and portfolio choices fixed. Suppose the constraint remains binding and both investor types continue to hold both bonds.

At any given government price, the higher requirement makes domestic investors purchase more government bonds. With bond supply and the foreign demand schedule unchanged, this increase in domestic demand raises the equilibrium price for government debt. Foreign investors purchase fewer bonds at the higher price, so the domestic ownership share rises.

The domestic budget and supranational optimality condition also imply a reduction in domestic holdings of supranational bonds. Foreign investors absorb the residual supranational supply at a lower price. Higher government prices and lower supranational prices then compress the DBP. Conditional on unchanged default and relative liquidity compensation, the decomposition in (20) implies a larger financial repression wedge.

Under these conditions, tighter regulation produces both greater domestic government ownership and a larger financial repression wedge. This motivates using domestic ownership as an observable proxy for captive demand. Nonetheless, ownership also responds to foreign intermediation capacity and bond supplies, so it does not uniquely identify repression. The empirical analysis therefore examines its association with the DBP conditional on default risk, relative liquidity, real interest rates, bond supplies, and other market controls.

# 3 Data

**Supranational bond data.** To estimate the DBP and assess its sensitivity to the components derived in Section 2, I construct a novel dataset of supranational zero-coupon yield curves for 11 EM currencies. These yield curves are built from security-level data covering secondary market prices and characteristics of bonds issued by eight major supranational institutions. I obtain the data from Bloomberg Terminal, and restrict the bond universe to fixed-rate and zero-coupon securities that have no embedded options. Therefore, inflation-linked, floating-rate, callable,

and putable bonds are excluded from my estimation sample.

To be able to estimate reliable yield curves, one needs a sufficient number of secondary market quotes on each trading day. To address this challenge, in each currency, I pool bonds issued by the following agencies: the International Bank for Reconstruction and Development (IBRD), International Finance Corporation (IFC), European Bank for Reconstruction and Development (EBRD), European Investment Bank (EIB), Asian Development Bank (ADB), Kreditanstalt für Wiederaufbau (KfW), Inter-American Development Bank (IADB), and the African Development Bank (AfDB). These institutions are AAA-rated throughout my sample period and constitute the largest issuers of supranational bonds in EM currencies, ensuring that the resulting yield curves do not mix together bonds of heterogeneous credit quality.

The final sample of 4,017 securities provides a near-complete coverage of the AAA-rated supranational bond market in EM currencies. Table 1 details the issuance patterns by currency and issuer.

Table 1: Supranational bond issuance by currency and issuer (USD millions)

| | EIB | IBRD | EBRD | IFC | ADB | KfW | IADB | AfDB | **All** |
|---|---|---|---|---|---|---|---|---|---|
| TRY | 23,725 | 6,931 | 19,044 | 7,455 | 7,216 | 6,263 | 1,538 | 1,826 | **73,998** |
| ZAR | 13,785 | 18,211 | 10,953 | 2,663 | 6,171 | 5,097 | 1,269 | 6,846 | **64,997** |
| BRL | 8,342 | 11,339 | 7,769 | 8,414 | 3,704 | 5,501 | 2,765 | 1,542 | **49,374** |
| CNY | 1,081 | 5,006 | 1,907 | 3,198 | 5,504 | 7,480 | 0 | 1,424 | **25,601** |
| MXN | 3,442 | 6,712 | 722 | 5,624 | 975 | 886 | 2,053 | 988 | **21,402** |
| INR | 737 | 5,829 | 5,364 | 3,056 | 1,462 | 32 | 3,882 | 326 | **20,687** |
| PLN | 14,625 | 1,129 | 697 | 7 | 940 | 802 | 0 | 0 | **18,199** |
| RUB | 2,262 | 3,236 | 3,577 | 2,737 | 468 | 691 | 48 | 455 | **13,476** |
| IDR | 2,234 | 1,415 | 5,040 | 131 | 158 | 133 | 3,051 | 229 | **12,391** |
| HUF | 2,059 | 97 | 241 | 292 | 263 | 386 | 0 | 0 | **3,338** |
| COP | 0 | 1,149 | 118 | 591 | 995 | 0 | 74 | 0 | **2,927** |
| **All** | **72,293** | **61,055** | **55,432** | **34,168** | **27,856** | **27,271** | **14,679** | **13,637** | **306,392** |

*Notes:* This table reports the aggregate amount issued by an issuer in a given currency. Units are in millions of US dollars. The sample contains 4,017 supranational bonds issued between 1997 and 2025 across 11 EM currencies. BRL = Brazilian real; CNY = Chinese renminbi; COP = Colombian peso; HUF = Hungarian forint; IDR = Indonesian rupiah; INR = Indian rupee; MXN = Mexican peso; PLN = Polish zloty; RUB = Russian rouble; TRY = Turkish lira; ZAR = South African rand. Sources: Bloomberg and author's calculations.

**Supranational yield curves.** Computing the DBP requires extracting supranational zero-coupon yields that are comparable to government yields at fixed tenors. I construct these yields using a Dynamic Nelson-Siegel term-structure model (see Diebold and Li, 2006; Diebold et al., 2006), which is a standard method in the empirical asset pricing literature. The technical details of the estimation procedure are presented in Appendix E.

**Government yield curves.** Unlike supranational bonds in EM currencies, government yield curves are readily available from commercial data providers. For each currency, I obtain the comparable government zero-coupon yields at 1, 3, 5, and 10-year tenors directly from Bloomberg. To ensure a complete time-series for the entire sample period, I impute missing observations using equivalent series from Refinitiv. Details on other data sources are presented in Appendix C.

# 4 Summary statistics

Table 2 presents summary statistics for government and supranational zero-coupon yields and the DBP across the 11 EM currencies in my sample. Government and supranational yields are broadly similar in levels, but the DBP widens with maturity. Government bonds typically yield below supranationals at shorter maturities, with a median DBP of −28 basis points at the one-year tenor, and above them at longer maturities, with a median of +32 basis points at the ten-year tenor.

Table 2: Summary statistics of EM LC zero-coupon yields and the DBP

| | | Tenor | | | |
|---|---|---|---|---|---|
| | | 1-year | 3-year | 5-year | 10-year |
| Median | Supranational | 6.853 | 6.888 | 7.014 | 7.250 |
| | Government | 6.590 | 6.947 | 7.263 | 7.644 |
| | DBP | −0.284 | −0.003 | 0.123 | 0.321 |
| Std. dev. | Supranational | 6.700 | 5.326 | 4.500 | 3.662 |
| | Government | 5.348 | 4.538 | 4.021 | 3.354 |
| | DBP | 3.260 | 2.555 | 2.118 | 1.723 |
| Minimum | Supranational | 0.101 | 0.110 | 0.433 | −0.103 |
| | Government | −0.024 | 0.107 | 0.429 | 1.178 |
| | DBP | −40.500 | −33.006 | −26.127 | −19.630 |
| Maximum | Supranational | 57.261 | 49.070 | 41.297 | 32.168 |
| | Government | 50.621 | 43.018 | 34.753 | 25.784 |
| | DBP | 4.911 | 5.529 | 6.133 | 7.836 |

*Notes:* This table reports the median, standard deviation, minimum, and maximum values for the zero-coupon yields and the DBP across 11 EM currencies throughout the sample period (2007-2025). Units for yields are in percent and DBP is in percentage points. Sources: Bloomberg, Refinitiv, and author's calculations.

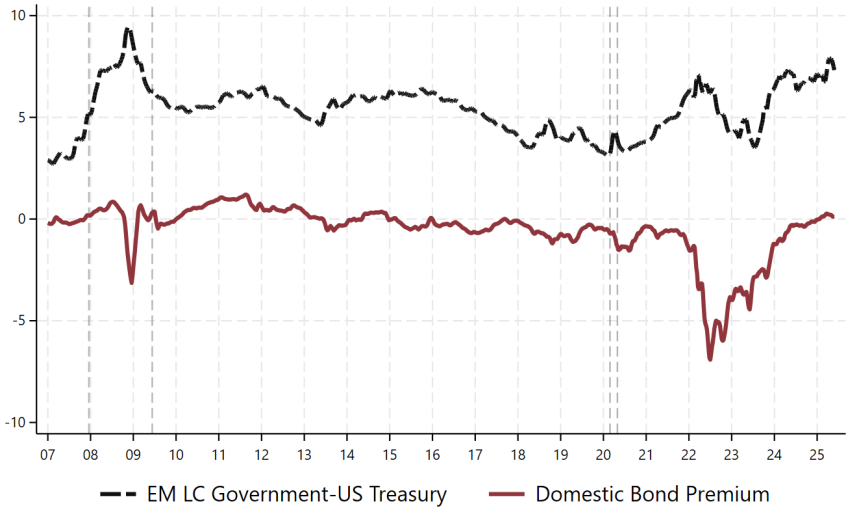
The DBP also varies substantially across countries and over time. Its standard deviation ranges from 1.7 to 3.3 percentage points, and one-year observations range from approximately −41 to +5 percentage points. The largest negative observations occur during episodes of severe market stress in Russia and Turkey, examined in Section 5.5.

Figure 2 compares cross-country averages of the DBP and the EM LC government–US Treasury yield spread at one- and five-year maturities, both for the full sample and excluding Russia and Turkey. The DBP is generally much smaller than the spread over US Treasuries, since the former excludes FX risk premia by construction.

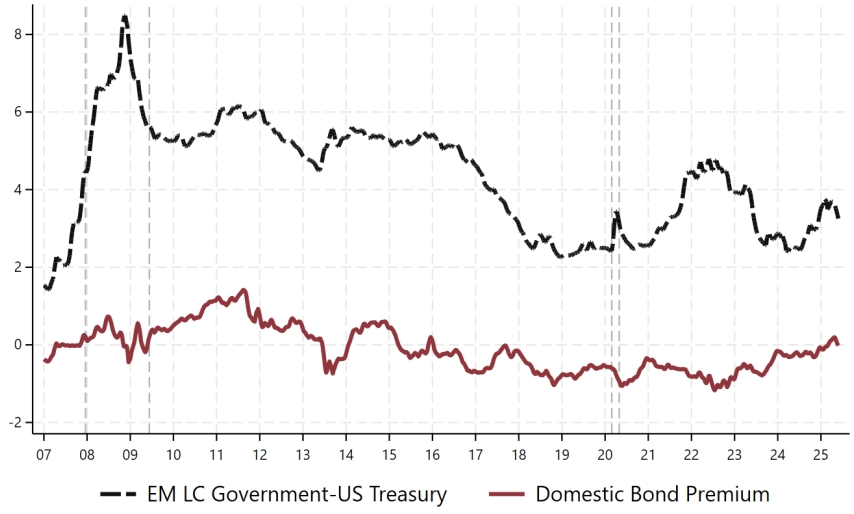
The average DBP declines over much of the sample. At the five-year tenor, it falls from around 150 basis points in 2012 to near zero by 2025. There are, however, substantial deviations from this trend. The full-sample averages show particularly large negative spreads in 2022–2023, followed by a sharp reversal. These movements are less pronounced when Russia and Turkey are excluded, although at shorter tenors, the DBP is still negative over much of the sample period.

The two spreads also behave differently during times of financial stress. During the Global Financial Crisis and the pandemic, EM LC government–US Treasury spreads widened sharply.

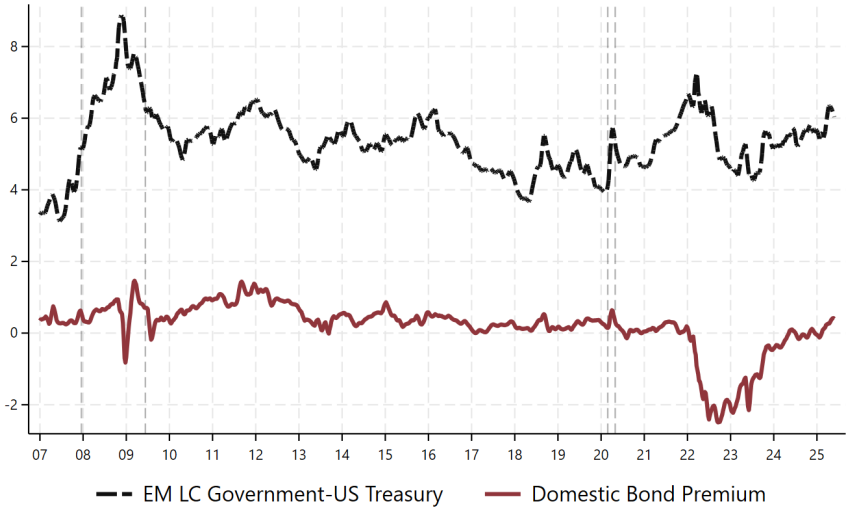
Figure 2: Cross-country averages of the EM LC government–US Treasury spread and the DBP



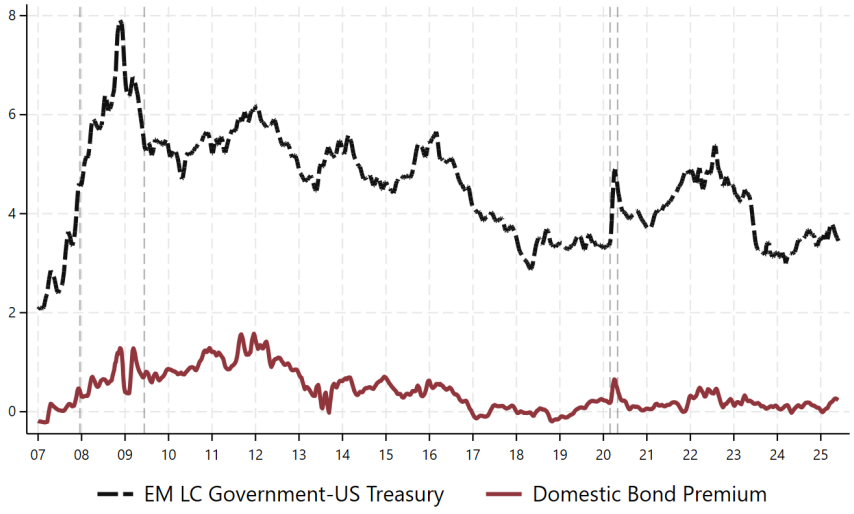

(a) One-year tenor (full sample)




(b) One-year tenor (ex-Russia & Turkey)




(c) Five-year tenor (full sample)




(d) Five-year tenor (ex-Russia & Turkey)

*Notes:* This figure shows the cross-country, four-week rolling averages of the EM LC government–US Treasury spread (black) and the DBP (red) at weekly frequency. Units are in percentage points. The full sample comprises BRL, CNY, COP, HUF, IDR, INR, MXN, PLN, RUB, TRY, and ZAR; the ex-Russia & Turkey panels exclude RUB and TRY. Sources: Bloomberg, Refinitiv, and author's calculations.

By contrast, DBPs declined or at least they were much more stable. This is particularly noteworthy because compensation for expected sovereign haircuts should raise government yields relative to supranational bonds in crises. These patterns therefore motivate examining the role of financial repression as a potential explanation. The next section aims to do this by testing the predictions of the model presented in Section 2 and estimating the sensitivity of DBP to sovereign default risk, illiquidity, and financial repression.

# 5 What explains the domestic bond premium?

## 5.1 Global versus local factors

If financial repression is an important factor in EM LC government bond pricing, one would expect the DBP variation to be predominantly country-specific, as repressive policies respond to domestic fiscal pressures. To test this prediction, I decompose the overall DBP variance into components attributable to persistent country characteristics, common global shocks, and idiosyncratic local factors (Appendix F provides further details on the methodology).

Table 3: Variance decomposition of the DBP

| **Tenor** | **Between-variance** | **Within-variance** | |
|---|---|---|---|
| | | Local factors | Global factors |
| 1-year | 11.73% | 69.32% | 18.95% |
| 3-year | 11.59% | 72.22% | 16.19% |
| 5-year | 14.17% | 72.84% | 12.99% |
| 10-year | 20.61% | 68.68% | 10.71% |

*Notes*: This table reports the variance decomposition of the DBP panel during the sample period (2007–2025). Data aggregated to monthly frequency prior to decomposition. See Appendix F for further details on the methodology. Sources: Bloomberg, Refinitiv, and author's calculations.

The decompositions in Table 3 show that, at each maturity length, around 70% of the overall variance in DBP can be explained by idiosyncratic and time-varying (i.e., local) factors. Meanwhile, common shocks that hit all markets simultaneously (global factors) account for only 10-19% of the variance. The dominance of idiosyncratic factors contrasts with Longstaff et al. (2011), who document a large common component in sovereign CDS spreads across countries.

The stark difference reflects the fact that DBP removes the FX risk premiums, which are highly correlated across EMs. Indeed, a principal component (PC) analysis on EM exchange rates in my sample between 2007 and 2025 vis-à-vis the US dollar reveals that the first PC explains 81% of the total variance in exchange rate movements. This large common component suggests limited risk-sharing across EM currencies.

Furthermore, persistent cross-country differences (between variance) explain at most 21% of the DBP variance, suggesting that time-varying factors rather than fixed country characteristics account for most of the variation in government borrowing costs relative to default-free benchmarks. Overall, these patterns are consistent with local macro-financial conditions or domestic policies, including financial repression, driving the yield differentials.[7]

[7]However, the decomposition should be interpreted conservatively, as global shocks may interact with local economic conditions and time-varying factors can eventually influence persistent country characteristics.

The percentages in Table 3 also provide upper bounds on what different sets of covariates can achieve. Global time-varying factors *alone* cannot explain more than 19% of the DBP variation, and fixed cross-sectional controls are capped at around 20%. As such, my empirical strategy relies on currency-tenor fixed effects and time-varying controls, with an emphasis on understanding the contribution of time-varying variables that proxy for the DBP components.

### 5.2 Testing the equilibrium predictions of the theoretical framework

This section tests the predictions of the framework presented in Section 2 by assessing the sensitivity of DBP to each component. I map the model-implied decomposition (20) to a panel regression that controls for sovereign default risk with CDS spreads[8], the relative liquidity term $\widehat{L}^{GS}_{j,t,\tau}$ with government and supranational bond bid-ask spreads, and absorbs persistent cross-sectional differences in the DBP with currency–tenor fixed effects.

The theoretical framework predicts that tighter financial regulation can produce both greater domestic ownership and a larger repression premium conditional on default and relative liquidity compensation. I therefore use the domestic ownership share of government bonds as a proxy for the intensity of financial repression in panel regressions.

However, financial repression need not operate solely through captive domestic demand. Inflationary monetary policy accompanied by regulatory policies that trap domestic savers can also compress borrowing costs, so lower real policy rates often go hand-in-hand with financial repression (Reinhart and Sbrancia, 2015; Kose et al., 2022). I therefore include the real policy rate as an additional control. I measure it using central bank policy rate instead of government bond yields for two reasons. First, government bond yields are an input to the DBP and using them would introduce mechanical correlation. Second, central bank rates better reflect the intended monetary policy stance, whereas government bond yields may also incorporate factors such as financial intermediaries' external funding costs (e.g., De Leo et al., 2022).

The resulting specification is:

$$\begin{aligned} DBP_{j,t,\tau} = \alpha + \beta_1 CDS_{j,t} + \beta_2 BidAsk^{Gov}_{j,t,\tau} + \beta_3 BidAsk^{Sup}_{j,t,\tau} + \beta_4 Dom.\ Holdings_{j,t-1} \\ + \beta_5 Real\ Policy\ Rate_{j,t-1} + \gamma' X_{j,t,\tau} + \phi_{j,\tau} + \nu_{j,t,\tau}, \end{aligned} \tag{25}$$

where $Dom.\ Holdings_{j,t-1}$ is the lagged level of the domestic ownership share of LC government bonds. $Real\ Policy\ Rate_{j,t-1}$ captures the lagged real policy rate. The control vector $X_{j,t,\tau}$ includes the lagged tenor-matched amounts outstanding of government and supranational bonds in all specifications; since bond supplies enter the equilibrium wedge in (23), these controls absorb supply-driven variation in the DBP. Domestic and global factors are added progressively, as indicated in Table 4. Based on the decomposition (20), I expect $\beta_4 < 0$ and $\beta_5 > 0$: higher domestic ownership of LC government bonds and lower real policy rates should be followed

[8] I use the five-year sovereign CDS spread, the most liquid contract with the broadest cross-country coverage. Section 5.3 checks robustness to using tenor-matched CDS-implied default probabilities. Crucially, although CDS contracts predominantly reference foreign currency obligations, the mismatch with LC bonds is limited. The risk channel specific to LC debt, de facto default through inflation and currency depreciation, affects both legs of the DBP symmetrically and nets out of the spread by construction. The remaining component, outright default risk, is similar across denominations; the average gap between EM sovereigns' local and foreign currency credit ratings, as wide as three notches in the mid-1990s, has since closed almost entirely (Amstad et al., 2020).

by a smaller DBP. Moreover, higher sovereign default risk should be associated with a higher ($\beta_1 > 0$) and lower supranational liquidity with a smaller DBP ($\beta_3 < 0$).

The estimation sample includes all 11 EM currencies but excludes Russian rouble observations from February 2022 and Turkish lira observations from May 2022 onward, to ensure that the baseline estimates are not driven by extreme repression episodes in which the DBP reached −30 percentage points. These episodes are examined separately as case studies in Section 5.5. All regressors are standardised at the currency–tenor level to facilitate comparison across variables with different units. The resulting coefficients measure the change in DBP (in percentage points) associated with a one-standard-deviation increase in each regressor. Following Du and Schreger (2016) and Dao and Gourinchas (2026), I estimate the regression equation using an OLS fixed-effects estimator with Driscoll and Kraay (1998) heteroskedasticity and autocorrelation-adjusted standard errors.[9]

Table 4 reports the results. The financial repression proxies are statistically significant and stable across all specifications. The domestic holdings coefficient is negative and ranges from −0.20 to −0.31 at the one-year tenor and from −0.19 to −0.27 at longer maturities, indicating that a one-standard-deviation increase in the domestic ownership share is followed by a 19–31 basis point decline in the DBP. The real policy rate coefficient is positive throughout, ranging from 0.24 to 0.30 at the one-year tenor and from 0.17 to 0.19 at longer maturities.

Sovereign default risk exhibits a maturity divide. The CDS coefficient is positive, large, and significant at longer maturities (0.23–0.26), indicating that a one-standard-deviation increase in CDS spreads is associated with a 23–26 basis point rise in DBP. At the short-end of the yield curve, it is statistically indistinguishable from zero, consistent with the low probability of nominal default over a one-year horizon for the EMs in my sample.

Supranational bid–ask spreads retain their expected negative sign and are significant across all specifications at both maturities, with a stronger effect at the one-year tenor (−0.20 to −0.33) than at longer maturities (−0.12 to −0.17). Coefficients on government bond bid-ask spreads are insignificant at short maturities; at longer maturities they are weakly positive in some specifications but insignificant in the full specification.

The coefficients also reveal a term structure of financial repression, though the pattern differs across channels. The real policy rate sensitivity is about one and a half times as large at the one-year tenor as at longer maturities, consistent with central bank policy rates anchoring the short end of the yield curve more strongly than long-term rates. The domestic holdings coefficient, by contrast, does not exhibit a clear maturity gradient and is, if anything, slightly larger at the one-year tenor. The remainder of this section examines the robustness of these findings.

### 5.3 Robustness and alternative specifications

A potential concern with the baseline specification is that the persistent components absorbed by the currency–tenor fixed effects, such as market segmentation, may not remain constant over time, as sustained policy interventions gradually reshape market structure and investor

[9]These standard errors are more appropriate than currency-level clustering when there are relatively few currencies (small $N$) and a long sample period (large $T$).

Table 4: Domestic bond premium, sovereign default risk, and financial repression

| | $DBP_{j,t,\tau}$ | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 1−year | | | | $>1$−year | | | |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
| $CDS_{j,t}$ | 0.042 | 0.005 | 0.011 | −0.021 | 0.262*** | 0.251*** | 0.234*** | 0.238*** |
| | (0.108) | (0.115) | (0.121) | (0.117) | (0.032) | (0.032) | (0.035) | (0.044) |
| $BidAsk^{Gov}_{j,t,\tau}$ | −0.006 | 0.019 | −0.028 | −0.051 | 0.027 | 0.057** | 0.073** | 0.026 |
| | (0.053) | (0.053) | (0.045) | (0.043) | (0.028) | (0.027) | (0.030) | (0.029) |
| $BidAsk^{Sup}_{j,t,\tau}$ | −0.331*** | −0.322*** | −0.236*** | −0.203*** | −0.135*** | −0.124*** | −0.170*** | −0.145*** |
| | (0.078) | (0.075) | (0.072) | (0.063) | (0.046) | (0.044) | (0.048) | (0.049) |
| $Dom.\ Holdings_{j,t-1}$ | −0.314*** | −0.279*** | −0.221*** | −0.199*** | −0.245*** | −0.224*** | −0.267*** | −0.186*** |
| | (0.088) | (0.077) | (0.067) | (0.070) | (0.056) | (0.052) | (0.058) | (0.053) |
| $Real\ Policy\ Rate_{j,t-1}$ | | 0.266*** | 0.239*** | 0.297*** | | 0.172*** | 0.173*** | 0.194*** |
| | | (0.076) | (0.069) | (0.075) | | (0.034) | (0.033) | (0.047) |
| *Controls* | | | | | | | | |
| Outstanding amounts | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Domestic factors | | | ✓ | ✓ | | | ✓ | ✓ |
| Global factors | | | | ✓ | | | | ✓ |
| Observations | 1,531 | 1,518 | 1,518 | 1,518 | 4,759 | 4,731 | 4,731 | 4,731 |
| Within-$R^2$ | 0.170 | 0.213 | 0.268 | 0.324 | 0.102 | 0.128 | 0.153 | 0.189 |

*Notes:* This table reports panel regression results of the DBP on proxies for default risk, illiquidity, and financial repression consistent with the model in Section 2. Estimation sample excludes RUB from February 2022 and TRY from May 2022 onward. Data in monthly frequency. All regressors are standardised at the currency–tenor level. $Dom.\ Holdings_{j,t-1}$ is the lagged level of the domestic ownership share of LC government bonds. $BidAsk^{Gov}_{j,t,\tau}$ and $BidAsk^{Sup}_{j,t,\tau}$ are bid–ask spreads for government and supranational bonds. Outstanding amounts: lagged tenor-matched government and supranational bond amounts outstanding. Domestic controls: Debt/GDP, FX volatility, domestic stock index returns. Global controls: VIX, broad dollar index (EM), US Treasury yields. All specifications include currency–tenor fixed effects. Driscoll and Kraay (1998) heteroskedasticity and autocorrelation robust standard errors in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Sources: Bloomberg, Refinitiv, BIS, IMF, and author's calculations.

behaviour. If these components drift, currency–tenor fixed effects may be insufficient and the independent variables could spuriously capture this slow-moving variation.

To address this, I replace currency–tenor fixed effects with currency-tenor-year fixed effects, which allow these persistent components to shift each year. This absorbs all common annual variation within each currency–tenor pair, including any slow-moving drift in these components, so that the financial repression proxies can only be significant if they explain within-year, within-currency-tenor movements in the DBP.

Table 5: Does captive demand insulate government borrowing costs from default risk pricing?

| | $DBP_{j,t,\tau}$ | | | |
|---|---|---|---|---|
| | 1−year | | > 1−year | |
| | (1) | (2) | (3) | (4) |
| $CDS_{j,t}$ | −0.079 | | 0.314*** | |
| | (0.099) | | (0.100) | |
| $CDS_{j,t} \times Dom.\ Holdings_{j,t-1}$ | −0.136* | | −0.119*** | |
| | (0.080) | | (0.025) | |
| $Default\ Prob._{j,t,\tau}$ | | −0.064 | | 0.131*** |
| | | (0.064) | | (0.047) |
| $Default\ Prob._{j,t,\tau} \times Dom.\ Holdings_{j,t-1}$ | | −0.115** | | −0.100*** |
| | | (0.048) | | (0.035) |
| $Dom.\ Holdings_{j,t-1}$ | −0.408*** | −0.430*** | −0.177*** | −0.148*** |
| | (0.106) | (0.134) | (0.063) | (0.057) |
| $Real\ Policy\ Rate_{j,t-1}$ | 0.123** | 0.134*** | 0.061* | 0.062* |
| | (0.055) | (0.048) | (0.034) | (0.032) |
| *Controls* | | | | |
| Liquidity | ✓ | ✓ | ✓ | ✓ |
| Outstanding amounts | ✓ | ✓ | ✓ | ✓ |
| Domestic factors | ✓ | ✓ | ✓ | ✓ |
| Global factors | ✓ | ✓ | ✓ | ✓ |
| Currency×tenor×year FE | ✓ | ✓ | ✓ | ✓ |
| Observations | 1,518 | 1,291 | 4,731 | 4,383 |
| Within-$R^2$ | 0.803 | 0.800 | 0.748 | 0.799 |

*Notes:* This table reports panel regression results of the DBP on proxies for default risk, illiquidity, and financial repression. The interaction between lagged domestic holdings and default risk is included to assess whether captive domestic demand insulates government borrowing costs from deteriorating creditworthiness. Sample excludes RUB observations from February 2022 and TRY observations from May 2022 onward. All regressors are standardised at the currency–tenor level. $Default\ Prob._{j,t,\tau}$ is the tenor-matched default probability implied by the sovereign CDS. Liquidity controls: government and supranational bid–ask spreads. Outstanding amounts: lagged tenor-matched government and supranational bond amounts outstanding. Domestic controls: Debt/GDP, FX volatility, domestic stock index returns. Global controls: VIX, broad dollar index (EM), US Treasury yield. Driscoll and Kraay (1998) heteroskedasticity and autocorrelation robust standard errors in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Sources: Bloomberg, Refinitiv, BIS, IMF, and author's calculations.

Table 5 reports the results. Columns (1) and (3) use the five-year CDS spread as the default risk proxy, while columns (2) and (4) replace it with tenor-matched CDS-implied default probabilities as a robustness check. In all specifications, I augment the default risk proxy by

adding an interaction with lagged domestic ownership share. This tests whether the sensitivity of the DBP to default risk depends on the composition of the pre-determined domestic ownership share.

The domestic holdings coefficient remains negative and significant. It is −0.41 at the one-year tenor and −0.18 at longer maturities in the CDS specification. The results are similar when tenor-matched default probabilities are used. The coefficient on real policy rate is still positive, though it attenuates at longer maturities.

The interaction term between default risk proxies and domestic ownership ratio is negative and significant throughout. This is consistent with captive demand for public debt insulating borrowing costs from deteriorating creditworthiness. At mean domestic ownership, a one-standard-deviation increase in CDS spreads is associated with an increase in the longer-term DBP by 31 basis points; at one standard deviation above the mean, this number falls to approximately 20 basis points.

In summary, the survival of the financial repression proxies under this demanding specification and the significant interaction term between default risk and domestic ownership support the interpretation that DBP helps track policy-driven distortions in government bond yields.

Table 6: Reserve requirements and the domestic bond premium

| | $DBP_{j,t,\tau}$ | | | |
|---|---|---|---|---|
| | 1−year | | > 1−year | |
| | (1) | (2) | (3) | (4) |
| $Default\ Prob._{j,t,\tau}$ | −0.010 | 0.046 | 0.188*** | 0.224** |
| | (0.112) | (0.166) | (0.052) | (0.094) |
| $Reserve\ Requirement_{j,t-1}$ | −0.198** | −0.278*** | −0.220*** | −0.279*** |
| | (0.097) | (0.095) | (0.056) | (0.069) |
| $Real\ Policy\ Rate_{j,t-1}$ | 0.316*** | 0.288** | 0.287*** | 0.309*** |
| | (0.103) | (0.109) | (0.073) | (0.082) |
| *Controls* | | | | |
| Liquidity | ✓ | ✓ | ✓ | ✓ |
| Outstanding amounts | ✓ | ✓ | ✓ | ✓ |
| Domestic factors | ✓ | ✓ | ✓ | ✓ |
| Global factors | ✓ | | ✓ | |
| Year FE | | ✓ | | ✓ |
| Observations | 275 | 275 | 774 | 774 |
| Within-$R^2$ | 0.474 | 0.585 | 0.313 | 0.399 |

*Notes:* This table reports panel regression results of the DBP on proxies for default risk and financial repression. In contrast to Table 4, I use reserve requirements as an alternative variable to capture regulatory pressure on banks' balance sheets. Data in quarterly frequency. All regressors are standardised at the currency–tenor level. $Reserve\ Requirement_{j,t-1}$ is the lag of average reserve requirement ratio from Federico et al. (2014), updated through 2021Q3. $Default\ Prob._{j,t,\tau}$ is the tenor-matched default probability implied by the sovereign CDS. Liquidity controls: government and supranational bid–ask spreads. Outstanding amounts: lagged tenor-matched government and supranational bond amounts outstanding. Domestic controls: Debt/GDP, FX volatility, domestic stock index returns. Global controls: VIX, broad dollar index (EM), US Treasury yield. All specifications include currency–tenor fixed effects. Driscoll and Kraay (1998) heteroskedasticity and autocorrelation robust standard errors in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Sources: Bloomberg, Refinitiv, BIS, IMF, Federico et al. (2014), and author's calculations.

A remaining concern is that the domestic ownership share may partly reflect voluntary home bias or investor portfolio choices that move endogenously with yields. To mitigate this concern, Table 6 replaces it with the lagged average reserve requirement ratio from Federico et al. (2014). Higher reserve requirements can increase demand for government debt through various channels. In several countries, reserve requirements can be satisfied partly through holdings of government securities. More generally, even when reserves must be held as balances at the central bank, higher reserve requirements tighten banks' balance-sheet constraints and reduce the funds available for lending to productive sectors. In the remaining allocation, LC government bonds often receive preferred regulatory treatment. For example, they carry lower risk weights, qualify as high-quality liquid assets, and can be pledged in central-bank liquidity operations, including repo auctions. Banks may therefore tilt residual portfolios toward sovereign debt, raising captive demand. Moreover, if the reserve requirements on FX-deposits or assets are increased, banks may tilt their portfolios toward safe LC assets, government debt being the primary destination. The lagged average reserve ratio therefore provides a cleaner proxy for regulatory pressure on banks' balance sheets, mitigating concerns that the estimated relation simply reflects an investor response to market conditions. Across specifications, a one-standard-deviation increase in lagged reserve requirements is associated with a 20–28 basis point decline in the DBP.

The next section provides a complementary high-frequency test using discrete macroprudential policy events, and Section 5.5 presents two case studies in which financial repression policies decoupled government borrowing costs from creditworthiness.

### 5.4 Macroprudential policies and government borrowing costs

The panel regressions in Sections 5.2 and 5.3 document that the DBP comoves with proxies for financial repression, but they do not show whether the DBP responds at high frequency to policy actions that are predicted to induce captive demand for government debt. This section provides such a validation exercise.

To do so, I construct a daily event panel from the narrative entries in the IMF Integrated Macroprudential Policy Database (iMaPP) (Alam et al., 2019), which records macroprudential policy actions across 134 countries through end-2024. From the detailed policy descriptions underlying this dataset, I extract the announcement or implementation dates and use a large language model (Claude Sonnet 4.6) to classify each action. A structured system prompt requires the model to return pre-specified fields for each narrative entry: the announcement and effective dates, the focal policy action, whether the action is relevant to government bond demand, the predicted direction of the effect, and the transmission channel (Appendix G describes dataset construction in more detail). With this dataset, I estimate the conditional dynamic associations between macroprudential policies and the DBP using daily panel local projections following Jordà (2005).

Many of the policy actions in my database are standard macroprudential tools such as reserve requirements, liquidity coverage ratios, or FX restrictions, and need not be designed to extract resources from the financial sector or reduce sovereign funding costs. However, even when a prudential policy is motivated by financial stability concerns, it can still affect

sovereign borrowing conditions by reshaping financial institutions' balance sheet constraints and altering their portfolio choices. Therefore, when sovereign securities are the most liquid LC asset eligible to satisfy regulatory requirements, captive demand for government debt may increase even without an explicit fiscal objective.

The empirical results presented here relate closely to the theory laid out in Section 2. A demand-increasing event is represented in the model as a rise in $\kappa_t$, which raises the equilibrium repression wedge $\rho_t$ when the constraint binds. Reis (2021b) offers a supporting structural account in which macroprudential regulation is modelled as the share of government bonds that banks must hold, and tightening that requirement mechanically raises bond prices through a captive-demand channel. Holding the other components of decomposition (20) fixed, both formulations deliver the prediction that the DBP should fall following demand-increasing (tightening) events and rise following demand-decreasing (loosening) events. Macroprudential actions classified as not relevant for government bond demand, such as loan-to-value ratios, should leave the DBP unchanged; these neutral events therefore serve as a placebo.

To test these predictions, let $U_{j,t}$, $D_{j,t}$, and $N_{j,t}$ indicate demand-increasing, demand-decreasing, and demand-neutral macroprudential policy dates, respectively. I measure the response as the cumulative change in the DBP from the trading day before the event to horizon $h$. For each currency–tenor pair $(j, \tau)$ and horizon $h \in \{0, 1, \ldots, 60\}$ trading days, I estimate

$$\begin{aligned} DBP_{j,\tau,t+h} - DBP_{j,\tau,t-1} = {} & \beta_h^U U_{j,t} + \beta_h^D D_{j,t} + \beta_h^N N_{j,t} + \pi_h \Delta DBP_{j,\tau,t-1} \\ & + \theta_h^U U_{j,t-1} + \theta_h^D D_{j,t-1} + \theta_h^N N_{j,t-1} \\ & + \Gamma_h' X_{j,\tau,t-1} + \alpha_{j,\tau,y(t)} + \delta_t + \varepsilon_{j,\tau,t,h}, \end{aligned} \tag{26}$$

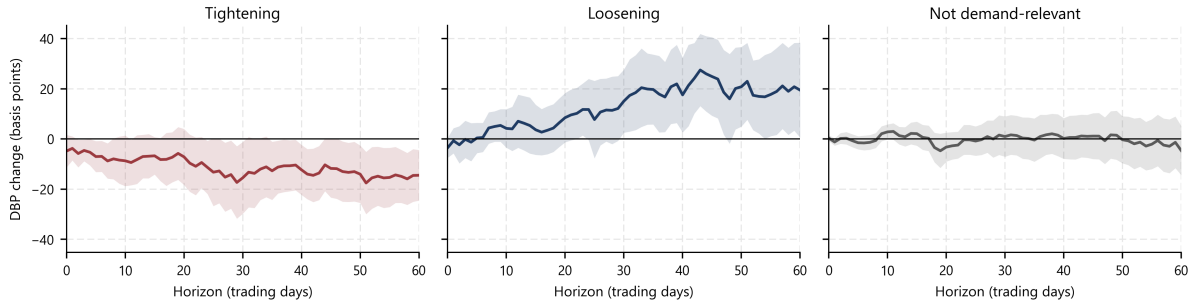
where $X_{j,\tau,t-1}$ is a vector of predetermined controls (the tenor-matched CDS-implied default probability and its 20-trading-day change, FX volatility, equity returns, the policy rate, and government and supranational bid–ask spreads), $\alpha_{j,\tau,y(t)}$ are currency–tenor–year fixed effects, and $\delta_t$ is a date fixed effect. The lagged change $\Delta DBP_{j,\tau,t-1}$ controls for short-run dynamics following Montiel Olea and Plagborg-Møller (2021). The fixed-effects absorb both slow-moving country and tenor-specific factors and global daily shocks.

Figure 3 plots the estimated response paths (the coefficients $\beta_h^U$, $\beta_h^D$, and $\beta_h^N$). On impact, the DBP is 5 basis points lower following a tightening event; the response builds gradually and reaches a trough of $-20$ basis points around 50 trading days after the event. Loosening events display a mirror-image pattern: a 15–25 basis point widening of the DBP over similar horizons. Both responses are statistically significant through most of the post-event window. Importantly, the estimated responses to neutral events hover around zero and are statistically insignificant at every horizon. The results are therefore consistent with the model's sign predictions.

Nonetheless, the timing of macroprudential actions is endogenous. Therefore, the goal of this exercise is more modest than identifying the causal impact of repression intensity on the DBP. Rather, it is to show that DBP co-moves with repressive policies at high frequency, in the direction predicted by the framework in Section 2.

The compression after demand-increasing events and the widening after demand-decreasing events support the interpretation of the DBP as a market-based measure that tracks policy-induced changes in domestic demand for government bonds. Moreover, the absence of a response

Figure 3: DBP response following macroprudential policy events




*Notes:* This figure plots the estimated cumulative response of the DBP over 60 trading days following tightening, loosening, and demand-neutral macroprudential policy events. Estimates use Kalman-filtered but unsmoothed supranational yield series to avoid look-ahead bias in local projections. Tightening events ($U$, left panel) are policy actions predicted to increase demand for government bonds; loosening events ($D$, centre panel) are actions predicted to decrease demand; neutral events ($N$, right panel) are macroprudential actions classified as not relevant for government bond demand. The dependent variable is $DBP_{j,\tau,t+h} - DBP_{j,\tau,t-1}$ for $h = 0, \ldots, 60$. Standard errors are Driscoll and Kraay (1998) with Bartlett kernel and horizon-dependent bandwidth $\max\{m(T), h+1\}$, where $m(T) = \lfloor 4(T/100)^{2/9} \rfloor$. Shaded regions are 90% confidence bands. Sources: Bloomberg, Refinitiv, BIS, IMF, and author's calculations.

to neutral events provides additional support for this interpretation. Finally, the pre-event diagnostic in Appendix G reduces the concern that these responses simply continue pre-existing DBP dynamics. Pre-event movements are statistically indistinguishable from zero at most horizons and, for tightening events, the point estimates, if anything, drift in the opposite direction. The post-event responses in Figure 3 therefore reflect movement that begins at the event date rather than a continuation of existing trends. I now turn to two case studies where financial repression mechanisms are especially visible.

## 5.5 Case studies

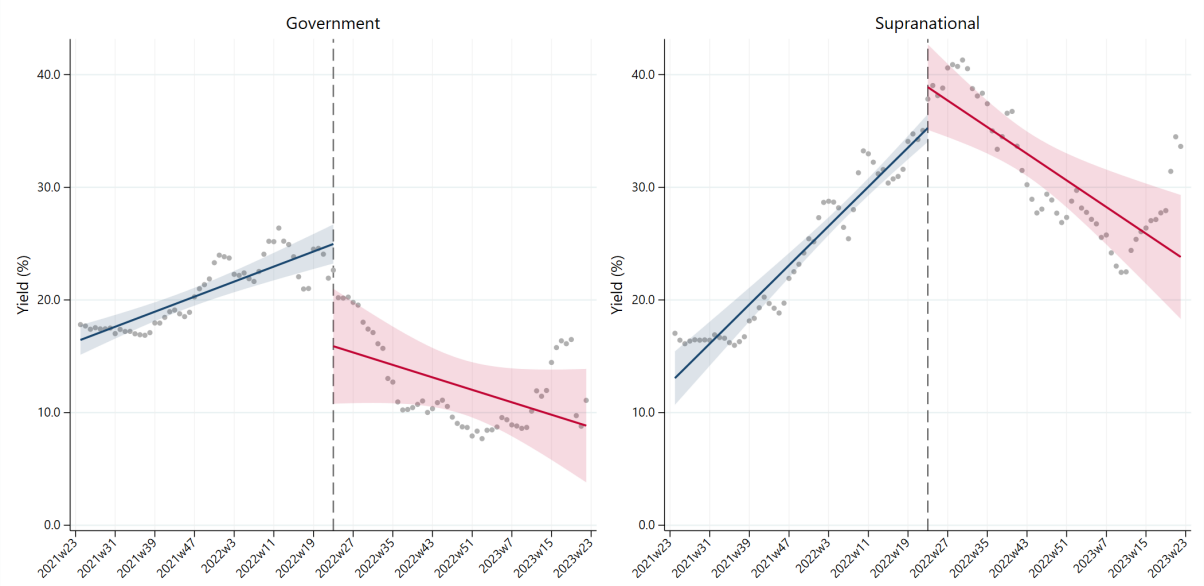
### 5.5.1 Turkey

Figure 1 in the introduction plots Turkish lira-denominated five-year government and supranational yields over the full sample. After the Global Financial Crisis and before the 2018 Turkish lira crisis, the five-year sovereign yield tracked the supranational one closely, reflecting a near-zero DBP.

Between 2019 and 2022, government yields gradually diverged below supranational yields, with the DBP turning negative and reaching approximately −5 percentage points by early 2022. This period was characterised by a sequence of policies designed to reduce dollarisation and raise domestic demand for LC government bonds: raising required reserve ratios for foreign currency deposits, imposing limits on banks' swap transactions with foreign counterparties, and applying moral suasion to reduce foreign currency lending. These measures pushed domestic institutions toward lira assets, with government bonds being the primary option. In 2021, authorities introduced exchange rate-protected lira deposit accounts, which put significant strain on government finances.[10] Meanwhile, the central bank faced increasing political pressure; between 2019 and 2023, the governor was replaced four times by the president and policy rates

[10]These foreign currency-linked accounts were guaranteed by the state and promised to pay the losses incurred on interest income due to exchange rate depreciations.

Figure 4: Five-year Turkish lira zero-coupon yields around the SMP announcement




*Notes:* This figure plots the weekly averages of daily five-year Turkish lira-denominated government (left) and supranational (right) zero-coupon yields, expressed in percent. The vertical dashed line marks the week containing 10 June 2022, when the detailed Securities Maintenance Practice requirements were published. The navy and red lines are separate pre- and post-event OLS linear fits, respectively; the event week enters both fits. Shaded regions are pointwise 95% confidence intervals for the fitted mean. These confidence bands are estimated using Newey–West heteroskedasticity-and-autocorrelation-robust standard errors. The sample covers the 51 weeks before and after the event week. Sources: Bloomberg, Refinitiv, and author's calculations.

were instructed to be kept low despite rising inflation (see Gürkaynak et al., 2023). During the same period, foreign participation in the LC government bond market declined by 15 percentage points, increasing the concentration of domestic holders.

These measures were followed by the most salient financial repression policy: the June 2022 announcement of the Securities Maintenance Practice (SMP).[11] This regulatory measure mandated banks to hold long-term LC government bonds in blocked accounts at the central bank, with requirements that linked their operational flexibility, including loan growth and interest rates, to their holdings of these securities (Celebi et al., 2025). Between mid-2022 and mid-2023, five-year nominal government yields declined by 20 percentage points despite 80% annual inflation and multiple credit rating downgrades. Following the gradual easing of the SMP requirements and monetary tightening after June 2023, yields reversed sharply, with government and supranational yields once again converging by 2024.[12]

Figure 4 plots the government and supranational yields around the official SMP announcement. Prior to June 2022, both yields rose, though supranational yields increased more steeply (from 17% to 34%) than government yields (from 18% to 24%), likely reflecting the asymmetric impact of policy interventions described above. The June 2022 announcement marks a significant divergence: government yields plummeted from approximately 24% to below 10% over the following months, while supranational yields spiked above 40% before gradually declining

[11] The SMP was informally announced at the May 26, 2022 Monetary Policy Committee meeting, with detailed requirements published in the official gazette on June 10, 2022. The policy took effect in July 2022.

[12] Beginning June 22, 2023, authorities initiated a gradual simplification of the SMP, reducing maintenance ratios over several months before fully terminating the policy on May 9, 2024 (Celebi et al., 2025).

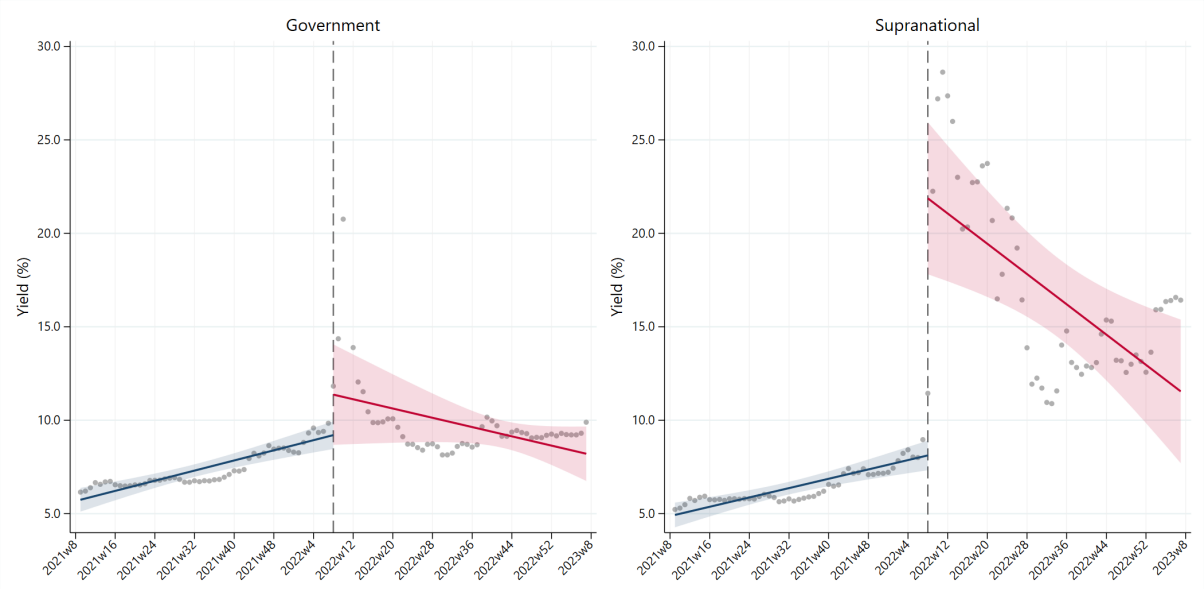
to around 30%.

While both yields co-move after the initial shock, they remain separated by approximately 20 percentage points through late 2023. This persistent gap coincides with the period when the SMP was in effect, suggesting the policy created additional captive domestic demand that helped decouple government yields from macroeconomic fundamentals. While concurrent monetary easing and deteriorating supranational liquidity prevent attributing the entire divergence to the SMP, these patterns are consistent with financial repression suppressing government borrowing costs below the unrepressed supranational benchmarks.

### 5.5.2 Russia

Figure 5 plots the Russian rouble-denominated five-year government and supranational yields around the beginning of the Russia-Ukraine war in February 2022. Prior to February 2022, both yields moved in tandem with government bonds trading slightly above supranational bonds (around 8-9% versus 7-8%, respectively). Following the war, government yields spiked to 20% before rapidly stabilising around 10%, while supranational yields surged to 30% and remained highly volatile around 15%, reversing the DBP from positive to deeply negative.

Figure 5: Five-year Russian rouble zero-coupon yields around the Russia-Ukraine war




*Notes:* This figure plots the weekly averages of daily five-year Russian rouble-denominated government (left) and supranational (right) zero-coupon yields, expressed in percent. The vertical dashed line marks the week containing 22 February 2022; Russia launched its full-scale invasion of Ukraine later that week. The navy and red lines are separate pre- and post-event OLS linear fits, respectively; the event week enters both fits. Shaded regions are pointwise 95% confidence intervals for the fitted mean. These confidence bands are estimated using Newey–West heteroskedasticity-and-autocorrelation-robust standard errors. The sample covers the 51 weeks before and after the event week. Sources: Bloomberg, Refinitiv, and author's calculations.

The initial government yield spike coincided with foreign investors exiting Russian LC government bonds (OFZs) – the nominal value of foreign holdings fell by 53% between January 2022 and June 2023. However, prices quickly stabilised as domestic financial institutions absorbed the excess supply in the secondary market. During this period, based on external debt statistics published by the Central Bank of Russia, domestic ownership share of OFZs reached 93%. With

international sanctions blocking access to global financial markets and capital controls imposed by the government restricting currency conversion, Russian financial institutions were pushed towards rouble-denominated assets, government bonds being the primary option.[13] Meanwhile, foreign investors largely withdrew from unsanctioned rouble-denominated supranational bonds amid broader sanctions on Russian assets. Russian investors could not actively trade these bonds either: sanctions severed their access to international clearing networks through which many of these bonds settled.[14]

The Russian case shows how an exogenous geopolitical shock can create captive demand for LC government debt. While supranational bonds languished without buyers, OFZs benefited from domestic institutions having fewer alternative investment options. The resulting negative DBP illustrates how this shock transformed domestic financial institutions into forced buyers of government debt.[15] Together, the Turkey and Russia examples illustrate how powerful financial repression can be in decoupling government borrowing costs from creditworthiness.

# 6 The fiscal footprint of financial repression

The preceding sections show that financial repression is visible in relative bond prices. Whether it delivers meaningful fiscal relief is a different question, since the answer depends not only on the size of the financial repression wedge but also on the quantity of debt that is associated with it. Multiplying the repression component of the DBP by the outstanding stock of nominal LC government bonds gives the additional amount the government would owe if that stock were refinanced at the unrepressed rates proxied by supranational yields.

In this section, I estimate a conservative lower bound for these savings, compare them with the revenue governments collect when inflation erodes the real value of their debt, a channel the DBP excludes by construction, and discuss what the magnitudes imply for repression as a policy tool for managing high public debt.

**Captive demand channel.** In the decomposition (20), a negative government-supranational spread reflects not only the repression wedge but also the relative liquidity term, and the regressions in Section 5.2 show that supranational bid-ask spreads are systematically associated with lower DBPs. Treating the entire negative spread as a saving from captive demand may therefore overstate the fiscal footprint. To purge this component, for each tenor I regress the DBP on default risk and liquidity proxies, lagged government and supranational amounts outstanding, and currency and time fixed effects, and subtract the fitted contribution of the supranational bid-ask spread from the raw DBP (Appendix H explains the procedure). Since government default risk and illiquidity may be correlated with repression, I do not subtract the fitted contribution of these components. However, increases in these components would push

[13] For information about international sanctions on Russian government debt, see Cleary Gottlieb Steen & Hamilton LLP (2023). For Russian capital controls, see Demertzis et al. (2022).

[14] See Kaniecki et al. (2024) on sanctions against Russian securities infrastructure including the National Settlement Depository.

[15] This pattern is consistent with Itskhoki and Mukhin (2025), who discuss that greater international financial isolation makes financial repression more potent in extracting fiscal surplus from the private sector.

the DBP up, therefore their omission from the adjustment only makes the savings estimates presented here more conservative.

For each currency $j$ and month $t$, I define the annualised fiscal saving as

$$Savings_{j,t} = \sum_{\tau} \max\left(0,\, -DBP^{adj}_{j,t,\tau}\right) B_{j,t,\tau}, \tag{27}$$

where $DBP^{adj}_{j,t,\tau}$ is the adjusted DBP and $B_{j,t,\tau}$ is the outstanding stock of nominal fixed-rate LC government bonds in tenor bucket $\tau$. Only observations with a negative adjusted DBP contribute. This asymmetry follows from the model, where the repression wedge satisfies $\rho_t \geq 0$. Because my bond sample excludes floating-rate and inflation-linked instruments, whose yields repression may also reduce, the estimates remain a lower bound and are best read as back-of-the-envelope calculations. The methodology follows the concept of debt revenue in Reis (2022), where the revenue a government collects from an interest rate discount equals the discount multiplied by the stock of liabilities that carries it.

Figure 6 reports the results. Panel (a) shows the cross-country mean and median of the savings from captive demand as a percentage of general government's overall revenue. The mean fluctuates around 0.15 percent of revenue and the median is mostly close to zero, meaning that in a typical month the typical government in my sample saves essentially nothing. The savings begin to rise around the Covid-19 crisis and peak in late 2022, when the mean reaches 0.65 percent of revenue and the median roughly 0.3 percent. In dollar terms, the sum across the eleven countries peaks at an annualised 48 billion US dollars in mid-2022.

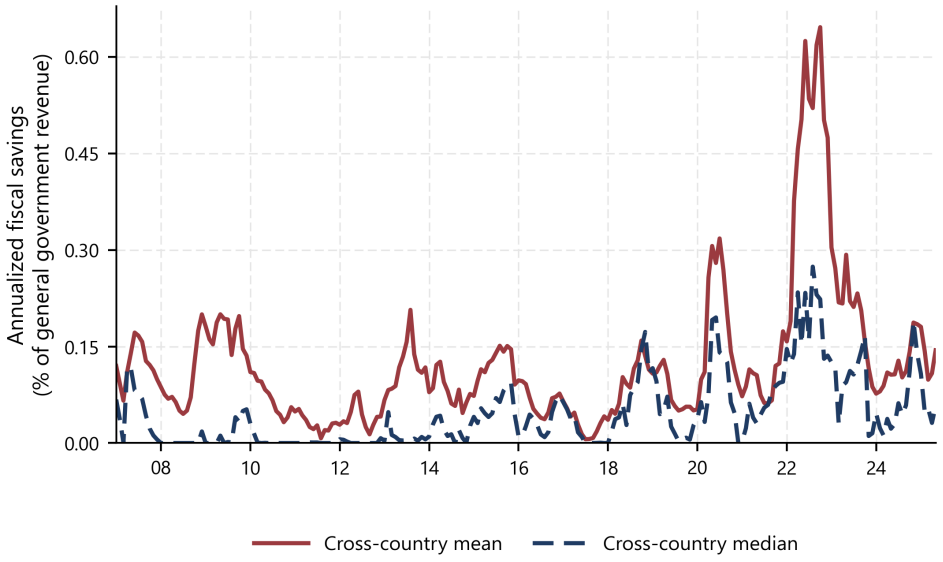
Panel (b) reports the weighted average across countries. The savings are negligible for most of the sample period. They began to rise after the pandemic, as public debt surged and financing conditions tightened. Yet even at its peak the aggregate saving stays below one percent of overall revenue. Therefore, over the past two decades, repression has been clearly visible in relative bond prices but second-order as a source of fiscal relief.

**Debt liquidation channel.** Giovannini and de Melo (1993) attribute roughly nine percent of government revenue to financial repression in developing economies during the 1970s and 1980s. Part of this substantial difference between my estimates and theirs is methodological. These earlier estimates benchmark domestic borrowing costs against foreign interest rates, so they also count the revenue governments collect when inflation erodes the real value of their debt.

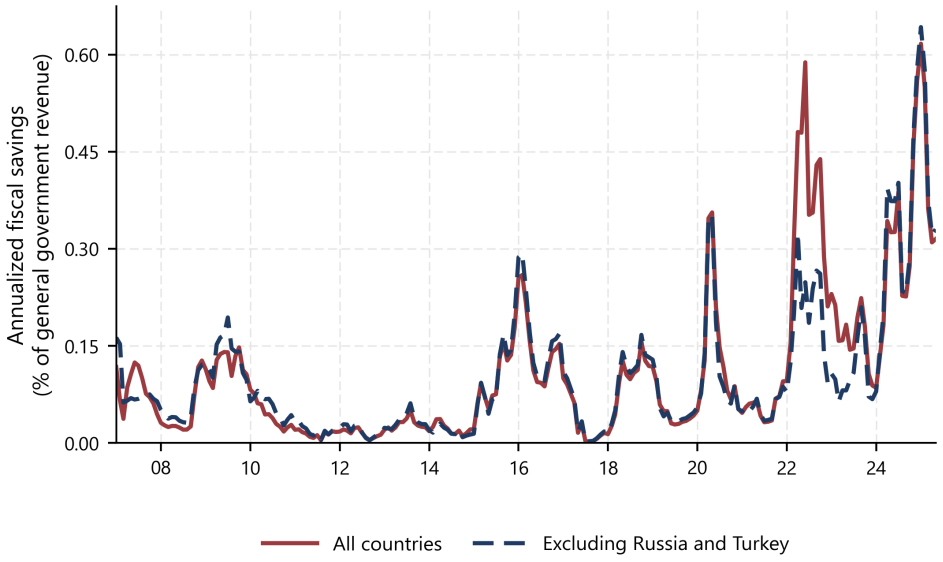
For fixed-rate bonds, when the nominal interest rates promised on government debt remain below realised inflation over the same period, the real value of the debt service payments erodes. Reinhart and Sbrancia (2015) call this the liquidation of government debt, and show that years of negative real returns, sustained by financial repression, produced the bulk of the fiscal consolidation in the postwar era.

Measuring this channel requires the contractual cost (i.e., yield at issue or coupon rates) of debt rather than market yields, since the revenue depends on the interest the government actually promised at the time of issuance. For each country $j$ and month $t$, let $i_{j,t}$ denote the weighted average contractual interest rate on the government bond portfolio, where weights are

Figure 6: Annualised fiscal savings from the captive-demand wedge




(a) Cross-country mean and median




(b) Cross-country weighted averages

*Notes:* This figure plots annualised fiscal savings from financial repression as a percentage of general government revenue. For each country and month, savings sum $\max(0, -DBP^{adj})$ times the outstanding stock of nominal fixed-rate LC government bonds across the four tenor buckets. Panel (a) shows the unweighted cross-country mean and median of the country ratios. Countries without an observation in a month are excluded from that month. Panel (b) divides total savings across countries by total government revenue across countries, for all eleven countries and excluding Russia and Turkey separately. The sample covers January 2007 to May 2025. Sources: Bloomberg, Refinitiv, IMF World Economic Outlook (April 2025), and author's calculations.

amounts outstanding of each fixed-rate bond. Over the year ending in month $t$, the government pays the rate locked in a year earlier, $i_{j,t-12}$, while realised twelve-month inflation $\pi_{j,t}$ erodes the real value of these payments. According to the Fisher equation, the ex post real rate is

$$r_{j,t} = \frac{1 + i_{j,t-12}}{1 + \pi_{j,t}} - 1, \tag{28}$$

and liquidation revenue equals the negative part of this real rate applied to the beginning-of-window nominal bond stock:

$$Liquidation_{j,t} = \max\left(0,\, -r_{j,t}\right) B_{j,t-12}, \tag{29}$$

where $B_{j,t-12}$ is the total outstanding stock of the fixed-rate and zero-coupon bonds, dated at the start of the window when the government's portfolio rate was locked in.[16] Both repression measures are thus annual flows on the same class of debt and are conceptually comparable.[17]

Figure 7 plots the cross-country mean and median of debt liquidation as a percentage of general government revenue. The typical country has no liquidation for most of the sample period. The mean stays below half a percent of overall government revenue for most of the time as well. This means that interest rates at issuance tend to exceed the inflation that follows. Still, the fiscal savings are large when the liquidation channel is active. As inflation surged after the pandemic while coupons on the outstanding stock remained fixed, the mean liquidation savings rose to 2.4 percent of government revenue in late 2022. This is more than three times the peak of the savings from captive demand shown in Figure 6. The fiscal savings then fell back toward zero by 2024 as new bonds were issued at higher interest rates. As such, for my sample of countries and time period, the transfers large enough to matter fiscally came through inflation rather than through the captive demand wedge alone.

Taken together, the results deliver a cautious message. Negative government-supranational spreads are common in the sample, across countries and over time. However, they are typically a few tens of basis points, and savings of that size do not map to large fiscal savings. Turkey shows what it takes for the revenue from financial repression to become economically significant. Even with borrowing costs held some twenty percentage points below the supranational benchmark by regulation that tied banks' operational flexibility to their government bond holdings, the savings amounted to around 2.6 percent of government's overall revenue in 2022.

The estimates are conservative, since the bond samples only cover fixed-rate nominal LC instruments; repression can also lower the interest rates on floating-rate, inflation-linked, and foreign currency debt. However, the conclusion is still difficult to avoid. Extracting sizable revenue from financial repression requires distorting bond markets and bank balance sheets to a degree that the revenue itself may be hard to justify.

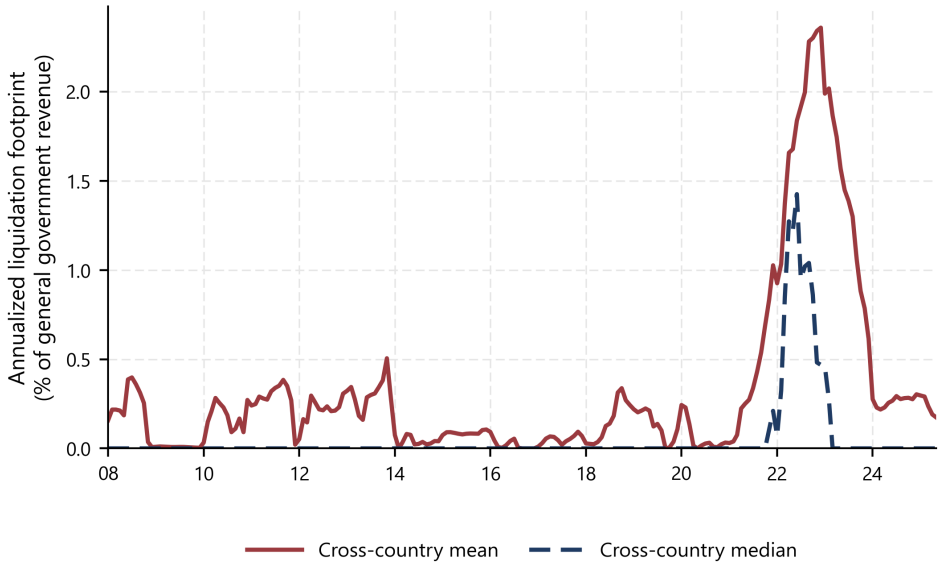
The two channels also work together rather than in isolation. Financial institutions required to hold government bonds while real returns are negative involuntarily subsidise the government. The cross-country means of the repression and liquidation savings co-move closely, with a monthly correlation of 0.76 that rises to 0.85 over twelve-month averages.

These magnitudes also explain why the postwar experience is different from today. The transfers of one to five percent of GDP per year that Reinhart and Sbrancia (2015) document required captive demand and negative real rate channels operating jointly and persistently. This may have been less difficult to achieve in financial systems where regulated interest rates and closed capital accounts left domestic savers few alternatives. The financial liberalisation that

[16] The zero cutoff for the real rate follows the convention of Reinhart and Sbrancia (2015)

[17] However, note that the two estimates are not additive. A coupon rate suppressed by captive demand at issuance lowers the contractual rate on this debt, and later reappears in the liquidation measure once inflation rises. Therefore, summing the two measures would double-count the savings.

Figure 7: Annualised debt liquidation through negative real interest rates




*Notes:* This figure plots the unweighted cross-country mean and median of annualised debt liquidation through negative real interest rates, in percent of general government revenue. For each country and month, the nominal portfolio rate is the outstanding-weighted average contractual interest rate on the stock of nominal fixed-rate LC government bonds. The real rate adjusts this by inflation realised over the following twelve months. The liquidation revenue equals the negative part of the real rate times the stock at the start of the window. Countries without an observation in a month are excluded from that month. Sample period between January 2008 and May 2025. Sources: Refinitiv, BIS, IMF World Economic Outlook (April 2025), national sources, and author's calculations.

reached EMs in the 1980s and 1990s changed this significantly (Abiad et al., 2010).

Currently, captive demand may be harder to sustain. Regulation can still push domestic institutions into government debt, but with significantly more open capital accounts, savings can move abroad when returns are artificially low. Foreign investors have also become a substantial presence in EM LC government bond markets (e.g., Onen et al., 2025). Because they sit beyond the reach of domestic regulation and are more price sensitive, their presence can limit the financial repression wedge. On the monetary side, inflation expectations in EMs have become better anchored over the past two decades, and anchored expectations stop temporary shocks from turning into persistent inflation (Bems et al., 2021). Negative real rate episodes therefore end within a few years rather than lasting decades. This gain in credibility is also related to the repression tax base itself. Governments can issue nominal fixed-rate LC debt to the extent that investors trust it will not be inflated away. Therefore, a government that relies on the inflation tax erodes the very tax base it aims to exploit.

A government can override these constraints with sufficiently strong interventions, but the costs of doing so are large. Compelling banks to hold large amounts of government debt can crowd out productive lending (Chari et al., 2020). Kliem et al. (2024) estimate that keeping postwar repression in place would have left US output five percent lower and the debt ratio twenty percentage points higher once general equilibrium effects are taken into account. More recently, Reis (2026) reviews the different financial repression channels available today and concludes that each delivers relatively limited revenue at a substantial efficiency cost. He also notes that persistent repression may push investors toward markets beyond the government's

reach, similar to US financial repression in the 1970s feeding the rapid growth of the offshore eurodollar market. Still, the temptation to conduct financial repression rises with the debt burden. Since the pandemic, with EM public debt at historically high levels, the fiscal footprint of repression has edged up. Whether it grows further as debt pressures mount is a question the DBP can track at high frequency.

# 7 Conclusion

This paper proposes the Domestic Bond Premium as a high-frequency measure to track financial repression and to quantify the revenue it generates. To compute it, I construct supranational zero-coupon yield curves for 11 EM currencies from a security-level panel of 4,017 bonds. Because the two legs of the spread share identical currency exposure, the DBP nets out the risk-free rate, leaving default risk, relative liquidity, and financial repression components.

Empirically, the DBP is persistently low and frequently negative, remains stable in crises when EM LC spreads over US Treasuries widen sharply, and is driven primarily by local rather than global factors. Financial repression proxies are significant in explaining its movements, and daily local projections around macroprudential policy events show the DBP moving in the directions the model predicts. The case studies of Turkey and Russia show repression holding government yields far below default-free benchmarks even as creditworthiness deteriorated. The fiscal revenue these policies generate, however, turns out to be modest. These findings have two broader implications.

First, financial repression can be tracked at daily frequency through asset prices. This complements subjective classifications and portfolio-based measures, which are available only at low frequency and with lags. Higher frequency and market-based measures can help policymakers and investors monitor the emergence of repression and its fiscal consequences in real time. Second, generating meaningful revenue through repression requires large price distortions. Savings from captive demand stay small even when the DBP is visibly negative, and become sizable only when borrowing costs are held far below the benchmark, as in Turkey, or when inflation erodes the real value of the debt. High levels of financial repression and inflation come with adverse macroeconomic consequences. Therefore, financial repression may be an unreliable tool to deal with rising public debt.

Several limitations temper these conclusions and point to future work. Supranational yield curves are estimated from bonds that trade at irregular maturities with varying liquidity, which may introduce measurement error. Regressions that validate the DBP as a measure that tracks financial repression remain correlational. Finally, causal identification of the macroeconomic consequences of financial repression would help quantify the trade-offs that policymakers face amid rising public debt.

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# Appendix

## A Model derivations

Appendix A derives the domestic investor's optimality conditions, bond yield decompositions, and the relation between financial repression intensity, domestic ownership share, and the DBP.

### A.1 Domestic optimality conditions

**Lagrangian.** Let $\mu_t$ and $\lambda_t$ be the multipliers on the budget and financial repression constraints. Using wealth $W_t$ from (2), the Lagrangian is

$$\begin{aligned}\mathcal{L}_0 = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \Big\{ & u(c_t) + v(T_{t+1}) \\ & + \mu_t \Big[ W_t - p_t c_t - \sum_{b \in \{Gov, Sup\}} (P_t^b + \varphi_t^b) B_{t+1}^{D,b} \Big] \\ & + \lambda_t \big[ P_t^{Gov} B_{t+1}^{D,Gov} - \kappa_t W_t \big] \Big\}.\end{aligned} \tag{30}$$

**First-order conditions.** Differentiating with respect to consumption gives

$$u'(c_t) - p_t \mu_t = 0 \quad \Longrightarrow \quad \mu_t = \frac{u'(c_t)}{p_t} > 0. \tag{31}$$

Differentiating with respect to supranational and government bond holdings gives, respectively,

$$\mu_t (P_t^{Sup} + \varphi_t^{Sup}) = v'(T_{t+1}) + \beta \mathbb{E}_t [\mu_{t+1} - \lambda_{t+1} \kappa_{t+1}], \tag{32}$$

$$\begin{aligned}\mu_t (P_t^{Gov} + \varphi_t^{Gov}) = & \, v'(T_{t+1}) + \lambda_t P_t^{Gov} \\ & + \beta \mathbb{E}_t [(\mu_{t+1} - \lambda_{t+1} \kappa_{t+1})(1 - \delta_{t+1})].\end{aligned} \tag{33}$$

Each bond provides private convenience services today. Upon maturity, it relaxes next period's budget constraint, but also raises the government-holdings requirement, which accounts for the term $\mu_{t+1} - \lambda_{t+1}\kappa_{t+1}$. Government bond holdings additionally relax the current regulatory constraint, hence the term $\lambda_t P_t^{Gov}$. Complementary slackness requires

$$\lambda_t \geq 0, \qquad P_t^{Gov} B_{t+1}^{D,Gov} \geq \kappa_t W_t, \qquad \lambda_t \big( P_t^{Gov} B_{t+1}^{D,Gov} - \kappa_t W_t \big) = 0. \tag{34}$$

The marginal value of nominal wealth is

$$\tilde{\mu}_t = \mu_t - \lambda_t \kappa_t. \tag{35}$$

The domestic value function is increasing and concave in wealth under the given preferences and constraints. Hence $\tilde{\mu}_t$ is positive and weakly decreasing in wealth. Dividing the bond first-order conditions by $\mu_t$ gives (7)–(8), with

$$\tilde{M}_{t+1} = \beta \frac{\tilde{\mu}_{t+1}}{\mu_t}, \qquad \rho_t = \frac{\lambda_t}{\mu_t}, \qquad \chi_t = \frac{v'(T_{t+1})}{\mu_t}. \tag{36}$$

## A.2 Bond prices and the DBP

Denote $\tilde{q}_t = \mathbb{E}_t[\tilde{M}_{t+1}]$ and $D_t = \mathbb{E}_t[\tilde{M}_{t+1}\delta_{t+1}]$ for the value of a unit of LC discounted under the constrained pricing kernel and discounted default loss, respectively. Rearranging the domestic Euler equations gives the price identities

$$P_t^{Sup} = \tilde{q}_t + \chi_t - \varphi_t^{Sup} \quad \text{and} \quad P_t^{Gov} = \frac{\tilde{q}_t - D_t + \chi_t - \varphi_t^{Gov}}{1 - \rho_t}, \tag{37}$$

where $0 \leq \rho_t < 1$. Taking logs of prices, multiplying by $-1$, and using the approximation $\log(1+x) \approx x$ around a frictionless, default-free benchmark gives the yield decompositions

$$y_t^{Sup} \approx \tilde{r}_t + \frac{\varphi_t^{Sup} - \chi_t}{\tilde{q}_t}, \tag{38}$$

$$y_t^{Gov} \approx \tilde{r}_t + \frac{D_t + \varphi_t^{Gov} - \chi_t}{\tilde{q}_t} - \rho_t, \tag{39}$$

where $\tilde{r}_t = -\log \tilde{q}_t$, which are the same as (16) and (17). Subtracting the supranational yield from the government gives the DBP decomposition

$$DBP_t = \log \frac{P_t^{Sup}}{P_t^{Gov}} \approx \frac{D_t}{\tilde{q}_t} + \frac{\varphi_t^{Gov} - \varphi_t^{Sup}}{\tilde{q}_t} - \rho_t. \tag{40}$$

The shadow risk free rates and private convenience terms cancel to first order, and $D_t/\tilde{q}_t = I_t^{Gov}$.

## A.3 Domestic ownership and a higher holdings requirement

The foreign investor's first-order condition for either bond is $V_t^{F,b} - P_t^b - \psi_t^b B_{t+1}^{F,b} = 0$. Combining it with market clearing gives

$$P_t^b = V_t^{F,b} - \psi_t^b\big(\bar{B}_t^b - B_{t+1}^{D,b}\big), \qquad b \in \{Gov, Sup\}. \tag{41}$$

For government bonds, this price and the domestic Euler equation give (23). Imposing the binding requirement $P_t^{Gov} B_{t+1}^{D,Gov} = \kappa_t W_t$ gives (24), which relates the domestic ownership share to the repression premium.

Consider a small increase in the current minimum government-holdings requirement $\kappa_t$, with no other change to the model's exogenous variables at any date. Hold current wealth and future bond prices in each possible state fixed, while allowing current consumption, bond prices and portfolios to adjust. Assume that repression remains binding with $\lambda_t > 0$ and that both investor types continue to hold both bonds.

Let $G = B_{t+1}^{D,Gov}$ and $S = B_{t+1}^{D,Sup}$, and omit current time subscripts. Differentiating the binding requirement $P^{Gov}(G)G = \kappa W$ and using $dP^{Gov}/dG = \psi^{Gov}$, gives

$$\frac{dG}{d\kappa} = \frac{W}{P^{Gov} + \psi^{Gov} G} > 0, \qquad \frac{dP^{Gov}}{d\kappa} = \psi^{Gov}\frac{dG}{d\kappa} > 0. \tag{42}$$

With government bond supply fixed, the domestic ownership share $s^D = G/\bar{B}^{Gov}$ also rises. The additional government purchases leave less resources for consumption and supranational

bonds:

$$pc = W - (P^{Gov}(G) + \varphi^{Gov})G - (P^{Sup}(S) + \varphi^{Sup})S. \tag{43}$$

Suppose $S$ remained unchanged or increased. Higher spending on bonds would reduce consumption and raise $\mu = u'(c)/p$. Since $P^{Sup}$ would not fall, the marginal utility cost of supranational purchases would rise. Their marginal benefit would fall: larger total holdings reduce $v'(G+S)$, while higher next-period wealth weakly reduces $\tilde{\mu}_{t+1}$. These changes violate the supranational Euler equation (32), which equates marginal cost and benefit. Domestic supranational holdings must therefore fall.

Foreign investors must hold more supranational bonds and will do so only at a lower price given their downward sloping demand curve. Government yields fall while supranational yields rise, reducing the DBP:

$$\frac{dDBP}{d\kappa} = \frac{1}{P^{Sup}}\frac{dP^{Sup}}{d\kappa} - \frac{1}{P^{Gov}}\frac{dP^{Gov}}{d\kappa} < 0. \tag{44}$$

As a result, controlling for default and relative liquidity compensation, the first-order decomposition (40) implies a higher repression wedge.

# B DBP vs LCCS: conceptual and empirical differences

This appendix compares the DBP with the local currency credit spread (LCCS) of Du and Schreger (2016) and argues the latter might confound domestic policy-induced distortions in government bond markets, as it also incorporates CIP deviations and the US Treasury premium.

## B.1 Currency-hedged US Treasuries as a benchmark

Fix currency $j$. Let $f_{j,t}$ denote the forward premium for converting dollars into currency $j$ next period (i.e., the cost of hedging the dollar payoff into currency $j$). The LCCS is defined as

$$LCCS_{j,\$,t} \equiv y^{Gov}_{j,t} - \left(y^{Gov}_{\$,t} + f_{j,t}\right), \tag{45}$$

where $y^{Gov}_{\$,t}$ is the U.S. Treasury zero-coupon yield. Then, $y^{Gov}_{\$,t} + f_{j,t}$ is the yield on a currency-hedged US Treasury, expressed in currency $j$ units. Using the yield decomposition of the model, the LCCS can be expressed as

$$LCCS_{j,\$,t} \approx \left(I^{Gov}_{j,t} - I^{Gov}_{\$,t}\right) + \left(L^{Gov}_{j,t} - L^{Gov}_{\$,t}\right) + \left(C^{Gov}_{j,t} - C^{Gov}_{\$,t}\right) - \left(\rho_{j,t} - \rho_{\$,t}\right) - CIP_{j,t}, \tag{46}$$

where the CIP basis is

$$CIP_{j,t,\tau} \equiv \left(\tilde{r}_{\$,t,\tau} + f_{j,t,\tau}\right) - \tilde{r}_{j,t,\tau}. \tag{47}$$

When $CIP_{j,t,\tau} = 0$, LCCS isolates the non-currency components. However, Dao and Gourinchas (2026) document persistent CIP deviations in EMs since the early 2000s. Even if currency hedging markets are frictionless, convenience yield on US Treasuries may confound EM issuer-specific components (Krishnamurthy and Vissing-Jorgensen, 2012; Du et al., 2018). Furthermore, financial repression premium on US Treasuries may also be non-zero. For these reasons, I focus

on DBP as the primary measure to track financial repression intensity in EMs.

### B.2 Empirical comparison

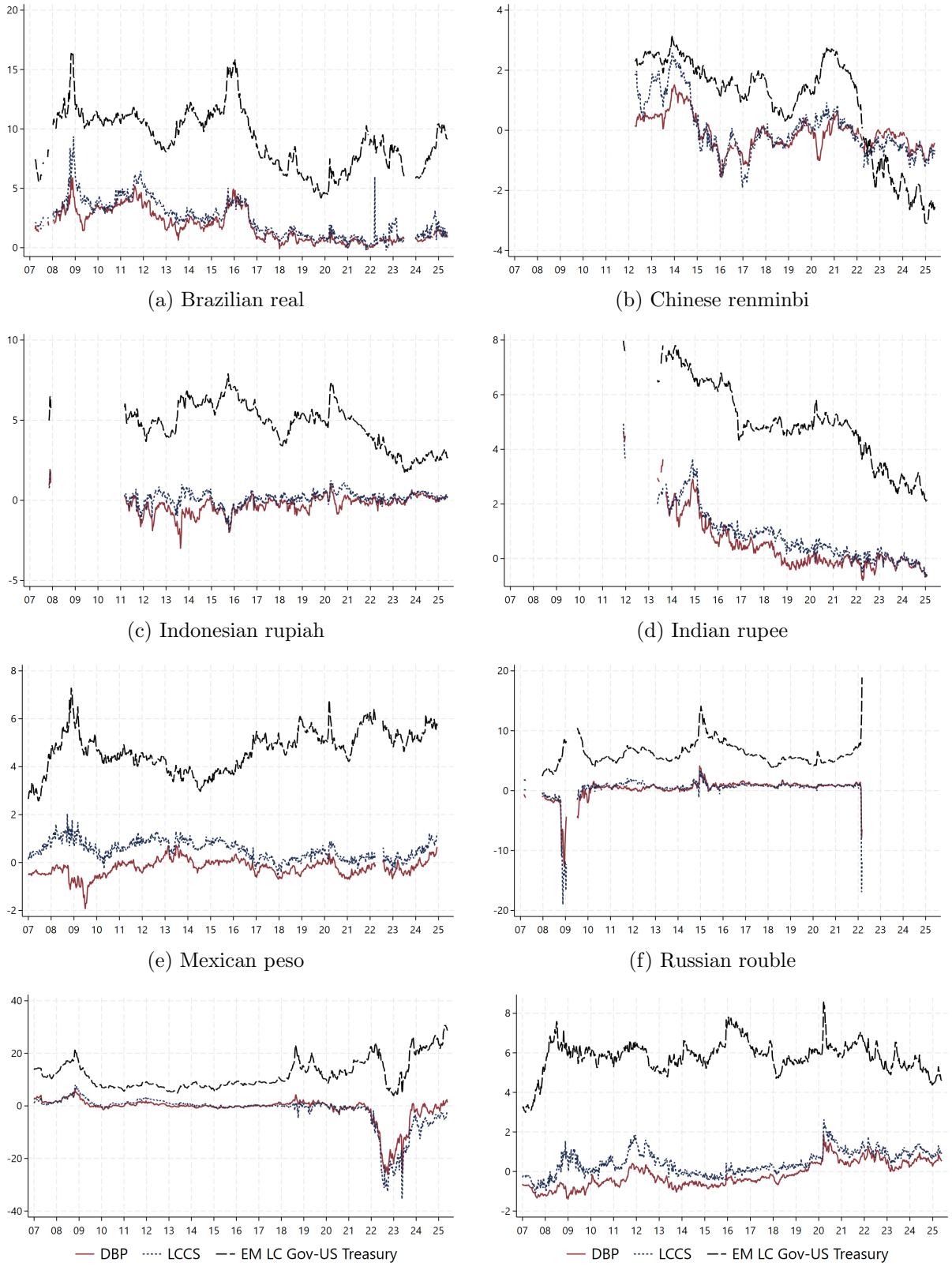
Figure 8 compares the five-year DBP with the LCCS defined in Appendix B.1. During the sample period when they are both available, the DBP and LCCS track each other closely (correlation $\approx 0.93$) and both lie well below the EM LC government–US Treasury spread, except in China after 2022, where policy rates were cut while other major economies tightened aggressively. This suggests that DBP and LCCS capture similar risk components.

However, the DBP tends to lie below the LCCS in most countries and time periods, and the two measures occasionally diverge during stress episodes (most notably in Brazil, South Africa, and Mexico during the GFC).

The LCCS embeds CIP deviations, US Treasury convenience yields, and a potential financial repression premium on US Treasuries, which can account for the LCCS lying modestly above the DBP even in normal times. These components widen during stress episodes: CIP deviations increase as financial intermediary balance sheets become more constrained (Du and Schreger, 2022), US Treasury yields decline relative to other developed market sovereigns as investors seek safe havens (Du et al., 2018), and captive domestic demand for Treasuries may rise, inflating the LCCS even if EM yields remain unchanged. Additionally, forward contracts for EM currencies vary in structure across countries and time (e.g., deliverable versus non-deliverable) and can become highly illiquid during financial stress, contributing further noise to the LCCS.

Naturally, the DBP could also be affected by the liquidity differentials between government and supranational bonds, but this is relatively straightforward to control in regressions using proxies for secondary-market liquidity. Therefore, the nature of measurement challenges in both spreads differs significantly: the LCCS may be influenced by global phenomena such as CIP deviations and US Treasury premium that affect all EMs simultaneously, while frictions embedded in the DBP tend to be idiosyncratic and amenable to direct measurement through secondary-market liquidity proxies.

Figure 8: Five-year yield spreads (DBP, LCCS, and EM LC government–US Treasury)




(a) Brazilian real

(b) Chinese renminbi

(c) Indonesian rupiah

(d) Indian rupee

(e) Mexican peso

(f) Russian rouble

(g) Turkish lira

(h) South African rand

*Notes:* This figure plots the DBP (solid red line), LCCS (dotted blue line), and EM local-currency government–US Treasury spread (dashed black line) at the five-year tenor, expressed in percentage points, for eight EM currencies. Each panel retains dates for which all three measures are available. Sources: Bloomberg, Refinitiv, Du and Schreger (2016) (updated through 2025), and author's calculations.

# C Data sources

Table 7: Variables and data sources

| Variable | Source |
|---|---|
| Supranational bond prices and characteristics | Bloomberg |
| Government zero-coupon yields | Bloomberg; Refinitiv |
| Government outstanding amounts | Refinitiv |
| Supranational outstanding amounts | Refinitiv |
| Contractual interest rates | Refinitiv |
| Sovereign CDS and default probabilities | Datastream |
| VIX, Broad Dollar Index | St. Louis Federal Reserve |
| Inflation rates | Datastream, Bank for International Settlements |
| Equity indices | Datastream |
| Spot exchange rates | Bank for International Settlements |
| Policy interest rates | Bank for International Settlements |
| EM currency forward premiums | Du and Schreger (2016) (updated, 2025) |
| Sovereign debt holdings; Debt-to-GDP | Arslanalp and Tsuda (2014) (updated, 2025) |
| General government revenue | IMF World Economic Outlook (April 2025) |

# D Government and supranational bond amounts outstanding

The empirical analysis uses the outstanding amounts of LC government and supranational bonds, as controls in the regressions of Section 5 and Appendix H, and as the debt stock in the fiscal footprint calculations of Section 6. I construct these series at monthly frequency from security-level issuance records. These records are sourced primarily from Refinitiv and supplemented by national sources. The bond sample consists of nominal fixed-rate LC bonds. I then aggregate the bonds into the 1, 3, 5, and 10-year tenor buckets. When constructing the series, I take into consideration re-openings, additional tranches, and buybacks of a given bond to the extent possible. Nonetheless, because the security-level records do not always capture every re-opening event, the resulting stocks are a lower bound of the fixed-rate total outstanding amount, which works to keep the savings estimates in Section 6 relatively conservative.

# E Term-structure model for supranational yield curves

**Bond-level price panel and preprocessing.** I estimate the supranational yield curves using a sample of 4,017 EM-currency denominated bonds. The bond sample is restricted to fixed-rate or zero-coupon/discount securities. Inflation-linked, floating-rate, callable, and putable securities are excluded from analysis. On each date, for each bond, I construct the schedule of remaining cash flows using the following information: issue date, maturity date, coupon rate, payment frequency, day-count convention, and end-of-month rule. I then convert the quoted clean-mid-price into a dirty price by adding the accrued interest.

Before estimating the term structure model, I apply several filters to remove observations that are unlikely to be informative about the cross-section of discount rates. First, I exclude January 1, December 25, and December 26, when trading activity is low. Second, I drop

observations with remaining maturity below 30 days, as well as rows with missing or non-positive clean prices. Third, I trim observations outside the 0.5th and 99.5th percentiles of the within-currency yield-to-maturity distribution. Fourth, I remove the dates on which the number of available quotes collapses relative to nearby dates within the same currency. Finally, I apply a filter that identifies erratic prices whose time-series behaviour is inconsistent with nearby observations for the same bond and cross-section behaviour with other similar bonds on the same date; flagged prices are interpolated when feasible and otherwise dropped. The resulting clean bond panel is then used to estimate a daily zero-coupon curve for each currency.

**Price-based Nelson-Siegel representation.** Let $\boldsymbol{\beta}_{j,t} \equiv \left[\beta^L_{j,t}\ \beta^S_{j,t}\ \beta^C_{j,t}\right]'$ denote the level, slope, and curvature factors for currency $j$ on date $t$. I model the continuously-compounded supranational zero-coupon rate at tenor $\tau$ using the standard Nelson-Siegel representation:

$$r^{Sup}_{j,t}(\tau) = \beta^L_{j,t} + \beta^S_{j,t}\left(\frac{1-e^{-\lambda\tau}}{\lambda\tau}\right) + \beta^C_{j,t}\left[\left(\frac{1-e^{-\lambda\tau}}{\lambda\tau}\right) - e^{-\lambda\tau}\right]. \tag{48}$$

To help with the stability of the maximum likelihood estimation, I fix the decay parameter at $\lambda = 0.7308$, which corresponds to the value set by Diebold and Li (2006).[18]

I write the measurement equation in price space. Let bond $i$ in currency $j$ have remaining cash flows $\{CF_{j,i,t,k}\}_{k=1}^{K_{j,i,t}}$ at times $\{s_{j,i,t,k}\}_{k=1}^{K_{j,i,t}}$, measured in years from the settlement date. The model-implied dirty price is

$$P^{dirty}_{j,i,t} = h_{j,i}\big(\boldsymbol{\beta}_{j,t}\big) + \varepsilon_{j,i,t} \equiv \sum_{k=1}^{K_{j,i,t}} CF_{j,i,t,k}\exp\big(-s_{j,i,t,k}\, r^{Sup}_{j,t}(s_{j,i,t,k})\big) + \varepsilon_{j,i,t}. \tag{49}$$

This formulation uses the full cash-flow structure of each coupon bond and avoids the need to first compress the raw data into synthetic yields at representative maturities.

**State dynamics and observation noise.** The latent factors evolve according to a mean-reverting diagonal AR(1) process:

$$\boldsymbol{\beta}_{j,t} = \boldsymbol{\mu}_j + A_j\big(\boldsymbol{\beta}_{j,t-1} - \boldsymbol{\mu}_j\big) + \boldsymbol{\eta}_{j,t}, \qquad A_j = \mathrm{diag}(\phi^L_j, \phi^S_j, \phi^C_j), \qquad \boldsymbol{\eta}_{j,t} \sim \mathcal{N}(\mathbf{0}, Q_j). \tag{50}$$

The covariance matrix $Q_j$ is unrestricted, allowing shocks to the three factors to be correlated. The diagonal transition matrix follows Diebold and Li (2006). The measurement errors are assumed conditionally Gaussian and diagonal in price space:

$$\varepsilon_{j,t} \sim \mathcal{N}(\mathbf{0}, R_{j,t}), \qquad R_{j,t} = \mathrm{diag}\big(\sigma^2_{j,1,t}, \ldots, \sigma^2_{j,N_{j,t},t}\big), \tag{51}$$

with

$$\sigma_{j,i,t} = \max\left\{\sigma_{p,j},\ \frac{1}{2}BAS_{j,i,t}\right\}, \tag{52}$$

[18] This choice of $\lambda$ places the curvature hump at roughly 30 months. The typical remaining maturity in my sample is around 2 years, consistent with the choice of $\lambda$.

where $\sigma_{p,j}$ is a currency-specific baseline price-noisiness parameter and $BAS_{j,i,t}$ is the observed bid–ask spread. This adjustment treats bonds with higher bid–ask spreads as noisier measurements of the true zero-yield curve. Because the measurement equation (49) is nonlinear in the factors, the filter takes the form of an extended Kalman filter.

**Initialisation, filtering, and smoothing.** I initialise the maximum likelihood estimation with a sequence of static cross-sectional Nelson-Siegel fits in the price space. Specifically, on each usable date, I solve

$$\widehat{\boldsymbol{\beta}}_{j,t}^{seed} = \arg\min_{\boldsymbol{\beta}} \sum_{i=1}^{N_{j,t}} \left[ P_{j,i,t}^{dirty} - h_{j,i}(\boldsymbol{\beta}) \right]^2. \tag{53}$$

These seed estimates are then placed on a business-day state grid running from the first to the last observed quote date, excluding January 1, December 25, and December 26. Missing weekdays are filled by time interpolation for initialisation. I then estimate the long-run mean $\boldsymbol{\mu}_j$ and the persistence parameters $(\phi_j^L, \phi_j^S, \phi_j^C)$ from factor-by-factor AR(1) regressions on the seed series, set $Q_j$ equal to the covariance matrix of the resulting transition residuals, and estimate the baseline price-noisiness parameter $\sigma_{p,j}$ from the static price residuals.

Given these starting values, I run an extended Kalman filter on the full business-day state grid and then apply a Rauch-Tung-Striebel smoother to obtain the final latent factor estimates.

**Reported zero-coupon yields.** After smoothing, I evaluate the Nelson-Siegel curve on a fixed maturity grid. The empirical analysis in the paper uses the 1, 3, 5, and 10-year points. Let $\widehat{r}_{j,t,\tau}^{Sup}$ denote the smoothed continuously-compounded zero rate implied by (48). The reported annualised zero-coupon yield is

$$\widehat{y}_{j,t,\tau}^{Sup} = \exp\left(\widehat{r}_{j,t,\tau}^{Sup}\right) - 1. \tag{54}$$

These supranational zero-coupon yields are then matched to the corresponding government zero-coupon yields to construct

$$DBP_{j,t,\tau} = y_{j,t,\tau}^{Gov} - \widehat{y}_{j,t,\tau}^{Sup}.$$

# F Variance decomposition methodology

The decomposition presented in Section 5.1 proceeds in two steps. First, I quantify how much of the spread variance reflects persistent cross-country differences attributable to slow-moving variables (e.g., institutional quality). These variables determine the average DBP in each currency. Consider a fixed tenor length $\tau$ and regress $DBP_{j,t,\tau}$ on currency fixed-effects $(\alpha_j)$ only:

$$DBP_{j,t,\tau} \;=\; \alpha_j \;+\; \varepsilon_{j,t,\tau}. \tag{55}$$

In this panel regression, the overall $R^2$ captures the between-variance $R_B^2$ – the share of variation explained by static currency/country characteristics. The complement $R_W^2 \equiv 1 - R_B^2$ represents the within-variance: the share of DBP variation explained by transitory variables such as fiscal

and monetary shocks, regulatory interventions, or evolving economic conditions.

Next, I distinguish between global factors that affect all currencies simultaneously and country-specific factors. To isolate time-varying components, I first demean the data at the currency-tenor level. Let $T_{j,\tau}$ denote the number of time-series observations in a currency-tenor pair. I compute currency-tenor means $\overline{DBP}_{j,\tau} = \frac{1}{T_{j,\tau}} \sum_t DBP_{j,t,\tau}$ and define the centred values as $\widetilde{DBP}_{j,t,\tau} = DBP_{j,t,\tau} - \overline{DBP}_{j,\tau}$. Regressing these centred values on time fixed effects ($\omega_t$) captures (average) shocks common to all currencies:

$$\widetilde{DBP}_{j,t,\tau} \;=\; \omega_t \;+\; \xi_{j,t,\tau}. \tag{56}$$

Let $R^2_{WG}$ be the overall $R^2$ from (56). Since the centred values in (56) contain only within-variation, the within-variance share can be decomposed as:

$$\text{Global factors: } R^2_G = R^2_W \times R^2_{WG} \quad \text{and} \quad \text{Local factors: } R^2_L = R^2_W \times \big(1 - R^2_{WG}\big),$$

so that $R^2_B + R^2_G + R^2_L = 100\%$ of the variance by construction.

# G Macroprudential policy events

## G.1 Data source

The iMaPP database (Alam et al., 2019) records macroprudential policy actions across 134 countries from 1990 onward, comprising both monthly tightening–loosening dummies and narrative descriptions of each action across 17 instrument categories. For the 11 sample currencies and the period from 2005 onward (to account for the events that may be announced before 2007 but implemented a few years later), I retain narrative cells for the parent instrument categories: reserve requirements (RR), liquidity-based measures (Liquidity), capital-based measures (Capital), restrictions on foreign-currency lending (LFC), limits on FX positions (LFX), restrictions on consumer or corporate loans (LoanR), leverage requirements (LVR), other macroprudential instruments (OT), and bank levies and taxes (Tax). This yields 767 policy descriptions.

## G.2 Classification

The structured extraction described in Section 5.4 requires the large language model to flag whether each policy action is relevant for government bond demand. For relevant actions, it then assigns a direction and a transmission channel. The direction label takes three values: $U$ for actions that increase demand for government bonds, $D$ for actions that decrease demand, and $A$ for relevant actions whose direction cannot be inferred from the text or that involve no substantive change (for example, a clarification). Events that are labelled as $A$ are emitted for completeness but do not enter the regression equation (26).

Actions classified as not relevant for government bond demand are typically prudential measures without a sovereign-specific dimension (e.g., leverage requirements or borrower-side loan restrictions). These actions form the neutral event set that enter equation (26) through the $N_{j,t}$ indicator.

To preserve an audit trail at the event level, I also require the model to quote the phrase from the source text justifying its direction and channel labels. From the 767 policy description cells, the classification yields 1,129 distinct focal actions, of which 602 are classified as relevant for government bond demand and 527 as not relevant.

### G.3 Classification prompt

The prompt I give to the large language model (Anthropic Claude Sonnet 4.6) is as follows.

```
You are a policy analyst extracting individual macroprudential policy actions
↪  from IMF iMaPP database narrative cells.

For each focal action, report:
- whether it is RELEVANT to equilibrium demand for local-currency (LC) government
↪  bonds,
- when relevant, the DIRECTION and CHANNEL of the effect,
- a short action summary and a reasoning quote from the text.

RELEVANT (boolean):
- true  - the action concerns equilibrium demand for LC government bonds, for
↪  example by mobilising captive domestic demand, by restricting
↪  foreign-investor access to the LC sovereign-bond market, or by restricting
↪  domestic access to foreign assets or non-sovereign alternatives.
- false - the action is outside this scope. Examples: pure household
↪  borrower-side rules, capital requirement restrictions without specific
↪  mention of sovereign treatment, generic loss-provisioning or accounting
↪  rules; framework-only publications; methodology notes.
Still emit the event when false, so we can use these actions as a placebo.

DIRECTION - populate ONLY when relevant=true; null otherwise.
- U (up): the action increases demand for sovereign debt. Examples: a new
↪  requirement that banks hold LC government securities through reserve or
↪  liquidity requirements; reserve-requirement hikes on FC deposits that push
↪  banks toward LC assets; risk-weight hikes on non-sovereign LC assets that
↪  make sovereigns relatively more attractive or risk-weight reductions on
↪  sovereign bonds; capital controls or FX-position caps that trap domestic
↪  savings or push foreign holders out, etc.
- D (down): the action decreases demand. Examples: termination or relaxation of a
↪  securities-holding mandate; suspension of a minimum LCR or NSFR (reduces HQLA
↪  demand for sovereigns); removal of capital controls on portfolio outflows;
↪  lowering a risk-weight OR taxation preferential treatment for sovereign
↪  holdings.
- A (ambiguous or neutral): a focal action whose net direction on LC sovereign
↪  bond demand cannot be signed - either because the text does not reveal
↪  direction, or because the action reaffirms / maintains / re-issues an
↪  existing rule with no substantive change (e.g., "reserve requirements were
↪  restructured"; "the Board confirmed the MPL at its current level").
```

```
CHANNEL - populate ONLY when relevant=true; null otherwise.
- mandate: a direct requirement that domestic institutions hold government
↪  securities. Examples: securities-maintenance rules; minimum LCR with
↪  sovereigns counted as HQLA; sovereign treatment in capital-adequacy
↪  frameworks.
- indirect: measures that narrow domestic institutions' non-sovereign or non
↪  local-currency denominated alternatives without an explicit sovereign bond
↪  quota. Examples: reserve requirements on LC or FC deposits or loans; limits
↪  on FC lending; FX-position caps on banks; risk-weight hikes on non-sovereign
↪  LC assets. etc.
- foreign: measures restricting foreign-investor access to the LC sovereign-bond
↪  market. Examples: capital controls on portfolio inflows or outflows;
↪  restrictions on non-residents' derivative transactions with domestic banks;
↪  currency-convertibility limits.
- other: clearly affects LC sovereign bond demand but does not fit any of the
↪  above. If you categorise it as such, explain what the mechanism could be.

Rules:
1. Emit one event per focal action. Read the cell holistically before splitting.
↪  "Date: M/D/YYYY" markers are dates, not necessarily event boundaries --
↪  multiple markers can describe aspects of the same action (one for announce,
↪  another for effective) or parts of a single policy package. Identify distinct
↪  focal actions first, then attach announce and effective dates from wherever
↪  they appear in the cell. When a cell opens with an announcement covering
↪  multiple related actions (a policy package -- same regulatory family,
↪  coordinated effective date), the opening announce date applies to all pieces
↪  of the package, not just the one explicitly named. Example: "Date: 6/10. It
↪  was announced RR will be raised, effective 6/24. Date: 6/24. Banks must also
↪  maintain additional securities." -- both pieces share announce=2022-06-10. A
↪  date or regulation mentioned in passing -- not as the cell's focus -- is
↪  context, not a separate event.
2. Emit every focal action, even if not repression-relevant. Set relevant=false
↪  for those, and leave direction and channel null. Do not silently drop
↪  non-relevant focal actions.
3. Use A sparingly - only when direction cannot be inferred (either the text does
↪  not reveal it, or the action is a non-change). Do not use A as a hedge when
↪  direction can be inferred from the wording.
4. Return announce_date and effective_date as YYYY-MM-DD ONLY when the text
↪  states a specific day. If the text gives only a month, quarter, or year,
↪  leave that field null. Do not infer day 01. Each date is evaluated
↪  independently. If the cell mentions the same action's date at multiple
↪  precisions (e.g., "announced in March 2013" and later "As announced March 3,
↪  2013"), use the most day-precise mention.
```

```
5. Reasoning must always be populated. When relevant=true, quote the phrase from
↪  the text that justifies BOTH the direction and the channel. When
↪  relevant=false, explain briefly why the action does not affect LC sovereign
↪  bond demand.
```

### G.4 Daily event panel

I construct the daily event panel used in equation (26) by retaining events with a precise announcement or effective date. I also drop any dates on which both $U$ and $D$ events appear. Relevant events with direction $U$ or $D$ supply the tightening and loosening indicators, and neutral events supply the $N_{j,t}$ indicator. Restricted to the 11 sample currencies and the period 2007–2024, the resulting panel contains 240 tightening and 97 loosening event-days. Table 8 summarises the daily event panel by country and direction.

### G.5 Pre-trend diagnostic

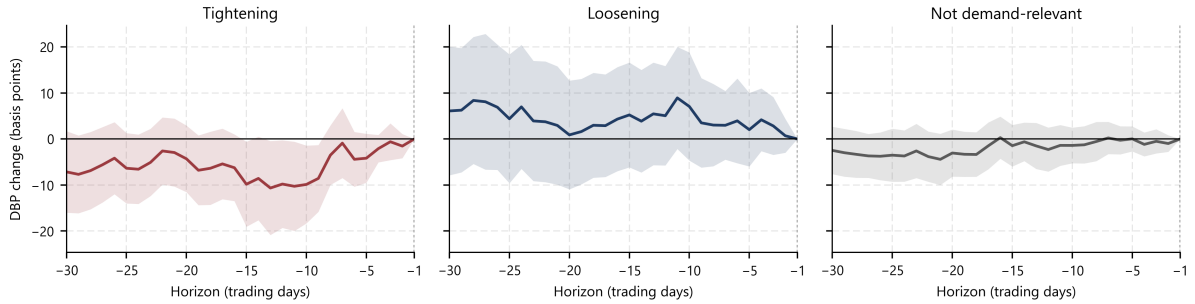
Figure 9 reports the pre-trends for the panel local-projection. To estimate these, I only use the contemporaneous treatment dummies and the currency–tenor–year and date fixed-effects of equation (26). I omit the lagged $\Delta DBP$, lagged treatment indicators, and lagged controls. For all event types, the pre-event coefficients are statistically indistinguishable from zero at most horizons. Where the tightening coefficients do differ from zero, they are mildly negative. This indicates that, on average, the DBP is slightly higher on the day before the event compared to the preceding two weeks. Therefore, it is not already drifting downward toward the post-event levels. The pre-trend diagnostic therefore supports the interpretation that the post-treatment trends in Figure 3 reflect movement starting at the treatment date rather than a continuation of pre-trends.

Table 8: Macroprudential policy events: country and direction

| Country | Tightening ($U$) | Loosening ($D$) | Ambiguous ($A$) | Total |
|---|---|---|---|---|
| Brazil | 11 | 18 | 3 | 32 |
| China | 45 | 19 | 0 | 64 |
| Colombia | 7 | 2 | 0 | 9 |
| Hungary | 16 | 2 | 3 | 21 |
| Indonesia | 19 | 7 | 2 | 28 |
| India | 22 | 11 | 0 | 33 |
| Mexico | 10 | 4 | 0 | 14 |
| Poland | 5 | 1 | 0 | 6 |
| Russia | 46 | 12 | 3 | 61 |
| Turkey | 52 | 20 | 0 | 72 |
| South Africa | 7 | 1 | 0 | 8 |
| **Total** | **240** | **97** | **11** | **348** |

*Notes:* Daily event-day counts from the collapsed event panel; restricted to the 11 sample currencies and 2007–2024. Underlying event-level channel mix among $U$ and $D$ events: mandate ($U = 134$, $D = 45$); indirect ($U = 264$, $D = 112$); foreign ($U = 7$, $D = 8$).

Figure 9: Pre-event dynamics around macroprudential policy events




*Notes:* This figure plots pre-event panel local-projection estimates of changes in the DBP, in basis points, from 30 trading days before a macroprudential policy event through the day before the event. Tightening events ($U$, red, left panel) are actions predicted to increase government-bond demand; loosening events ($D$, blue, centre panel) are actions predicted to decrease demand; demand-neutral events ($N$, black, right panel) are classified as unrelated to government-bond demand. The dependent variable is $DBP_{j,\tau,t+h} - DBP_{j,\tau,t-1}$ for $h = -30, \ldots, -1$, with $h = -1$ normalised to zero. The specification includes contemporaneous event indicators and currency–tenor–year and date fixed effects, but excludes lagged $\Delta DBP$, treatment indicators, and controls. Standard errors are Driscoll and Kraay (1998) with Bartlett kernel and horizon-dependent bandwidth $\max\{m(T), |h|+1\}$, where $m(T) = \lfloor 4(T/100)^{2/9} \rfloor$. Shaded regions are 90% confidence bands. Sources: Bloomberg, Refinitiv, IMF, and author's calculations.

# H Fiscal footprint estimates

This appendix describes the construction of the fiscal footprint measures presented in Section 6.

**Supranational illiquidity-adjustment.** For each tenor $\tau$ separately, I estimate a monthly panel regression of the DBP on the proxies for its non-repression components,

$$\begin{aligned} DBP_{j,t,\tau} = {} & \gamma_{1,\tau} CDS_{j,t} + \gamma_{2,\tau} BidAsk^{Gov}_{j,t,\tau} + \gamma_{3,\tau} BidAsk^{Sup}_{j,t,\tau} \\ & + \gamma_{4,\tau} Out^{Gov}_{j,t-1,\tau} + \gamma_{5,\tau} Out^{Sup}_{j,t-1,\tau} + \phi_j + \phi_t + \epsilon_{j,t,\tau}, \end{aligned} \tag{57}$$

where $Out^{Gov}_{j,t-1,\tau}$ and $Out^{Sup}_{j,t-1,\tau}$ are the lagged outstanding amounts of government and supranational bonds in the tenor bucket (in inverse hyperbolic sine), and $\phi_j$ and $\phi_t$ denote currency and month fixed effects. The estimation sample covers January 2007 to May 2025 and excludes Russia from February 2022 and Turkey from May 2022 onward, so that the coefficients are not driven by the two extreme episodes examined in the case studies. The estimated coefficients are then applied to the full panel. The adjusted DBP subtracts the fitted contribution of the supranational bid-ask spread,

$$DBP^{adj}_{j,t,\tau} = DBP_{j,t,\tau} - \hat{\gamma}_{3,\tau}\, BidAsk^{Sup}_{j,t,\tau}, \tag{58}$$

and thus corresponds to the DBP that would prevail if the measured supranational bid-ask spreads were zero. The remaining regressors serve as controls only, as default risk and government bid–ask spreads may themselves be related to repression, as discussed in Section 6.

**Using the adjusted DBP to measure fiscal savings.** The savings in equation (27) are computed for each currency and month on the outstanding stock of nominal fixed-rate LC government bonds in the four tenor buckets, valued in US dollars. Ratios use annual general

government revenue from the April 2025 IMF World Economic Outlook. Panel (a) of Figure 6 reports the unweighted cross-country mean and median of the country ratios, and panel (b) divides total savings across countries by the total revenue of the countries reporting in a given month. Each monthly observation is an annualised snapshot of the saving at prevailing spreads, and annual figures mentioned in the text are averages of the monthly snapshots.

**Contractual rates for the liquidation measure.** The contractual rate of a coupon-bearing bond is its coupon rate. For zero-coupon and discount bonds it is the yield at issuance. The nominal portfolio rate is the weighted average of the securities' contractual rates, where weights are the outstanding amounts of each bond divided by the total outstanding amount. The ex-post real rate and the liquidation revenue follow equations (28) and (29), with the portfolio rate and the bond stock fixed at the start of each twelve-month window. Inflation is measured by the realised change in consumer prices over the same window.